Chapter 6 — The Theory of Customer Understanding

How to discover what people are trying to accomplish, why they make the choices they make, and what evidence should change a founder's mind

Building What Lasts — Part II: Markets, Customers, and Differentiation

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Audience: Founders beginning from zero, product builders, researchers, educators, consultants, operators, and aspiring institutional leaders
Scope: Customer psychology, jobs-to-be-done, demand triggers, segmentation, behavioral economics, customer discovery, observation, interviews, purchasing processes, alternatives, problem urgency, evidence quality, and detailed applications to Bayes Institute.

> **Central thesis.** To understand a customer is not merely to know what they say they want. It is to construct a progressively testable explanation of **what they are trying to change, why it matters now, how they cope today, what alternatives they seriously consider, who can authorize the decision, what they must sacrifice, what prevents action, and what observable behavior would demonstrate genuine demand.** A good customer theory predicts decisions; a bad one only sounds empathetic.

A reader's map

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I — The Customer Is a Decision-Maker	Understand context, motivation, constraints, and behavioral psychology	1–12
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III — Acute Problems	Discover consequential, funded, time-sensitive struggles	22–30
IV — Segmentation	Group customers by decisions rather than decorative personas	31–38

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V — Field Discovery	Conduct interviews, observation, artifact research, and surveys	39–60
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How to use this chapter: Read Parts I–III for the core conceptual model; Parts IV–VI before interviewing or testing offers; Part VII before selling to organizations; and Part IX for the continuing Bayes Institute case. The appendices are intended for repeated use while conducting actual discovery.

Prologue — The customer who loved the idea and never bought

A founder speaks to thirty software engineers. Nearly everyone thinks AI matters. Most say that a disciplined, live, mentor-supported program would be useful. Several praise the syllabus. Some indicate they would pay a meaningful fee. The founder builds six months of instruction, hires reviewers, designs a polished website, and announces applications. Almost nobody enrolls.

There are many explanations. Perhaps the engineers sincerely valued learning but could not spare six months. Perhaps they already trusted their companies' internal support. Perhaps they feared being judged in an intensive cohort. Perhaps the actual purchaser would have been their employer. Perhaps their dominant alternative was free experimentation with AI tools, and its benefits were already sufficient. Perhaps the initial respondents were unusually supportive friends. Perhaps there was no immediate reason to act.

None of these explanations can be found by asking more people whether AI is important. That question never distinguished **interest from decision**.

Another founder investigates a different way. She asks each engineer about the *last real occasion* when they had to build something they did not know how to build. She reconstructs the task, deadline, consequences,

help sought, time spent, money paid, opinions consulted, alternatives discarded, and outcome. She observes an actual debugging session. She meets engineers who declined training. She meets a manager who sponsors learning but cannot approve external tools that access company code. She discovers that some engineers need a four-hour expert review of a live design, not a course; others need structured practice because they cannot sustain self-study; still others need credentials the new institution cannot provide. She makes one small paid offer to the subgroup whose actual behavior supports it.

Perhaps this second founder also fails. But she has begun to learn **which transaction exists**, rather than which idea sounds attractive.

Chapter 5 taught us that a market is a system of exchanges, not a headline statistic. Chapter 6 goes inside that system. It asks what causes a person or organization to switch from one state to another. The investigation is deliberately rigorous because almost every subsequent decision—product design, differentiation, price, distribution, sales, hiring, and expansion—depends on it.

Reading orientation and epistemic rules

The chapter progresses from decision theory to field research, from individual decisions to organizational purchases, and then to actionable customer evidence. The Bayes Institute case is an illustration of hypotheses and research design, **not** a claim that interviews, transactions, educational outcomes, institutional contracts, or a validated market already exist. Hypothetical numbers are worked examples, not survey findings or Indian market benchmarks. Research-derived theories are linked in the references; the canvases, matrices, rules of thumb, and proposed field procedures are managerial syntheses, not scientifically proven recipes. Psychology provides tendencies, not universal tricks for manipulating buyers.

Part I — The Customer Is a Decision-Maker, Not a Persona

1. The map is not the person

An ideal customer profile often reads like a passport: age 26, engineer, lives in a metro, earns an above-average salary, follows technology creators, wants professional growth. It looks specific. Yet it may not predict whether this person will buy a program next month. Two engineers with identical biographies can make opposite decisions because one has a pressing assignment and the other has a supportive workplace and no urgent skill gap.

The error is treating observable attributes as though they are the causes of action. Demographics can be useful for access, affordability, regulations, and descriptive comparisons. They often fail to explain *when and why* someone will accept a sacrifice. A founder wants a causal model: an incident occurs, the current approach becomes inadequate, a budget or opportunity opens, alternatives are evaluated, constraints are resolved, and one action wins.

This does not mean demographics are irrelevant, nor that people always narrate their motives accurately. The useful hierarchy is **context and observed behavior first; demographic attributes where they help explain exposure, constraints, access, or variation**.

2. The three questions hiding inside “Who is our customer?”

First, **who experiences the problem?** This is the user or beneficiary. Second, **who chooses the solution?** This is the selector, who may delegate research or rely on advisers. Third, **who pays and controls the commitment?** This is the economic buyer or budget holder. Sometimes all three are the same person. Often they are not.

In education, the learner may desire rigorous training, a parent may finance it, a company may sponsor it, and an employer may later interpret the result. In colleges, faculty may use the curriculum, students receive instruction, a department chair sponsors the change, procurement determines vendor eligibility, and finance releases funds. In B2B software, the end user can love the tool while the security officer vetoes it. A user interview cannot by itself establish purchasing authority.

The founder’s first customer diagram must therefore include **beneficiary, operator, champion, evaluator, economic buyer, approver, veto holder, and any gatekeeping intermediary**. Some actors play multiple roles. The diagram is incomplete until someone can say which person may sign, pay, cancel, or block the transaction.

3. A customer buys a change of state

People rarely want products for their own sake. They seek a difference between a present state and a desired one. A warehouse manager wants fewer stockouts, a parent wants reassurance, a commuter wants a tolerable journey, a developer wants to ship a reliable feature, and a learner wants to be capable of work that currently exceeds their skill.

But **desired change is not sufficient for a purchase**. The person must perceive a plausible path from purchase to that change, believe the expected gain exceeds monetary and nonmonetary sacrifice, and clear practical constraints. A company may desire better security yet never fund a security audit because a more immediate budget crisis absorbs its attention.

A useful, informal representation is:

$$ \text{Action} = f(\text{recognized gap},\text{stakes},\text{trigger},\text{alternatives},\text{trust},\text{resources},\text{friction},\text{authority}). $$

This is a causal checklist, not an estimable universal equation. It reminds us to look for conditions that make an action possible and timely rather than treating desire as a purchase order.

4. The need, the job, the desire, the request, and the purchase are different things

A **need** is an underlying condition or deficit. A **job** is the progress someone tries to make in a particular situation. A **desire** is a preferred experience or end state. A **feature request** is the customer's proposed mechanism. A **purchasing decision** is an actual exchange chosen against alternatives under constraints.

A learner says, "I need a course with thirty projects." Why thirty? Perhaps they want a credible portfolio; perhaps they mistake activity for mastery; perhaps they enjoyed project-heavy marketing. The intended job may be *gain independence in evaluating real AI systems*. Thirty projects are one proposed mechanism, perhaps an inefficient one. If you implement the request without identifying the job, you risk building a burdensome product that fails the customer's underlying goal.

Treat requests as hypotheses about mechanisms. Treat real decisions as evidence about trade-offs. Treat expressed aspirations respectfully, but do not confuse aspirations with immediate purchasing intent.

5. The four clocks inside every purchasing decision

A founder should distinguish four time horizons. **The problem clock** measures when the need emerges and when consequences grow. **The attention clock** measures when the buyer thinks about the problem and searches. **The budget clock** measures when money, authority, and procurement are available. **The delivery clock** measures when the proposed solution can create value.

A college may care intensely about applied AI electives but have fixed academic planning cycles; a corporate team may have an urgent deployment in six weeks but freeze vendor purchases until next quarter; a working engineer may have disposable income but no protected study hours. A technically excellent offer can fail because these clocks do not overlap.

An important early customer insight is often a better **buying occasion**, not a better product. Ask when the person becomes aware, when they decide, when money can move, and when results must arrive.

6. Customers are constrained optimizers, not fools to be persuaded

Standard economic models can illuminate trade-offs between benefits, prices, and alternatives. Herbert Simon's bounded-rationality work points to limits on cognition, search, and calculation [R1]. People use rules, habits, recommendations, and satisfactory defaults rather than analyzing every feasible choice.

A manager who selects an established vendor despite a cheaper startup may be rationally protecting uptime and career risk. A learner who postpones a promising course may be guarding time needed to keep their job. A shopper who buys a familiar product may be economizing on attention. Some decisions are biased; some simply reflect costs the founder forgot to count.

Never label refusal irrational before reconstructing the customer's constraints. A good entrepreneur studies those constraints and either solves them, changes the buyer or occasion, or accepts that the transaction should not happen.

7. What a customer knows and what a seller believes are different distributions

A founder experiences the labor, technical depth, and good intentions inside their offering. A buyer observes a description, price, visible evidence, referrals, and personal downside. This is an asymmetric-information problem. Akerlof's work on hidden quality explains why markets sometimes have trouble rewarding high-quality suppliers when buyers cannot reliably distinguish them [R2].

A new institute can be genuinely good at teaching but cannot expect prospects to act on the founder's private conviction. A potential customer might assign a low probability to successful delivery because many course advertisements overpromise. The founder must identify **which uncertainty matters**: instructor competence, learner fit, time commitment, credential recognition, program continuity, refund reliability, or real capability gain.

The point is not to manufacture trust by invoking psychological biases. It is to help the buyer reach a proportionate decision through transparent, claim-matched evidence.

8. People act relative to a reference point

The utility of an offer depends partly on what the buyer considers normal. Prospect theory emphasizes reference-dependent valuation and differing sensitivity to gains and losses [R3]. A founder can misunderstand this by announcing a feature as an improvement while customers see only the risk of losing what they already have.

Suppose a company replaces a familiar spreadsheet with software offering faster reporting. Users may value the saved hours, but also fear losing flexible manual control. Suppose a student considers a demanding program: the potential skill gain is attractive, but losing weekends is vivid and certain. The benefits may be delayed and uncertain.

Find the reference point before presenting value. What does the buyer already possess, understand, feel entitled to, and consider safe? What would adoption make them give up? Do not exaggerate threats or manufacture losses; understand the actual comparison they make.

9. Inertia is an active competitor

Samuelson and Zeckhauser documented status quo bias [R4], and research on retirement-plan defaults shows that choice architecture can substantially influence behavior in some settings [R5]. But in business, inaction also often has perfectly practical causes: implementation work, approval costs, sunk learning, contract commitments, uncertainty, and competing priorities.

When the incumbent seems good enough, a small improvement may not justify switching. A founder needs to learn the **switching threshold**: what change in performance, price, assurance, urgency, or support would make action worthwhile? Ask not merely why people dislike the incumbent, but why they *continue using it despite that dislike*.

“Everyone complains about the existing tool” is weak demand evidence if nobody has attempted to replace it. Persistent complaints can coexist with rationally high switching costs.

10. The purchase has a visible and an invisible price

The invoice is only one cost. Add **search time, onboarding, retraining, identity and status concerns, embarrassment, coordination, uncertainty, integration, study hours, travel, lost flexibility, managerial approval, and the risk of choosing wrongly**. The relative importance varies drastically between a biscuit and a six-month course.

For education, a ₹30,000 tuition fee with 150 hours of work is not simply a ₹30,000 decision. For an enterprise AI engagement, a ₹5 lakh contract might require far more internal staff time than the fee suggests. For FMCG, inconvenience and shelf absence may defeat an otherwise preferred brand.

A customer’s reported “too expensive” is ambiguous. It might mean cash shortage, insufficient value, weak belief, high time cost, or a cheaper trusted substitute. Do not reduce price before learning which sacrifice is decisive.

11. Trust, identity, and social risk can dominate functional features

People care about how a choice will be interpreted by colleagues, family, bosses, peers, and themselves. A young engineer buying an unproven course may worry about looking naive. A manager recommending an unknown vendor may worry about being blamed if implementation fails. A parent buying education may value institutional recognition because other people recognize it.

These social consequences are real even when they are hard to quantify. The customer may pay for confidence, belonging, recognized status, autonomy, or the ability to justify their choice. **Functional, emotional, and social jobs coexist**, and the mix is situation-dependent. Do not presume one universal psychological explanation for all education decisions.

12. What behavioral economics can and cannot tell founders

Behavioral research can alert founders to status quo bias, reference points, defaults, mental accounting, social influence, limited attention, present bias, and overconfidence. It rarely determines which tactic will work in a specific startup without customer-specific evidence. Experiments differ by setting and population; effect sizes do not transfer mechanically from retirement plans to enterprise procurement or education.

Use behavioral concepts as **diagnostic hypotheses**, not as a bag of “conversion hacks.” Ask what legitimate obstacle each concept might illuminate. An ethical product designer reduces confusion, makes trade-offs legible, helps customers make informed commitments, and avoids deceptive scarcity, fear, hidden charges, or manipulative defaults.

Part II — Jobs to Be Done: The Causal Anatomy of Demand

13. The job is the progress, not the product category

The jobs-to-be-done perspective associated with Clayton Christensen and colleagues emphasizes that customers choose products in particular circumstances to make progress [R6, R7]. A commuter might obtain a milkshake to make an otherwise tedious drive more tolerable; it competes in that occasion with things beyond other milkshakes. This example is an illustration of the lens, not evidence that every market is best modeled by the same story.

For a technical learner, "take an AI course" is an activity, not yet the job. Possible jobs include: *be capable of owning a production AI feature*, *avoid embarrassment when a manager asks difficult questions*, *prove competence before a role transition*, *build confidence through disciplined practice*, or *make time to learn despite competing obligations*.

The same institution may address several jobs, but the buyer rarely values them equally. Identifying the dominant job affects everything: curriculum depth, cadence, proof, admissions, payment, and distribution.

14. A useful job statement has a circumstance, struggle, and progress

Write: **When [specific situation], I struggle with [present obstacle] and want to [meaningful progress], while [important constraints].** Avoid naming your preferred product inside the job statement.

Weak: "When people want AI courses, they need personalized AI courses." This simply restates the product idea. Better: "When backend engineers are assigned a first customer-facing AI feature, they struggle to evaluate hallucinations, cost, latency, and failure modes; they want to make credible design decisions before the release review, while keeping up with their primary job."

The second statement reveals alternatives, deadlines, evidence, and product boundaries. It can lead to a course, technical review service, internal enterprise workshop, self-serve evaluation tool, or some combination. The right product remains an open question.

15. Functional, emotional, and social jobs

A functionally specified job might be "deploy a reliable forecasting system." Emotional progress might be "stop feeling out of control when executives ask why the forecast changed." Social progress might be "be perceived as a dependable technical leader." These dimensions may reinforce or conflict.

To discover them, ask about **consequences and reactions**: What would success change at work? What did failure feel like? Who would notice the result? What was embarrassing? What did the buyer want to avoid discussing? Respect boundaries; do not manufacture personal vulnerability to make a sale.

Emotional and social jobs are not always more important than functional jobs, nor are they evidence that customers are secretly irrational. They are components of the value being sought.

16. The four forces of progress

A useful JTBD-derived interview model asks about **push** from the current unsatisfactory situation, **pull** toward a new solution, **anxiety** about adopting it, and **habit** sustaining the old behavior. These four forces are qualitative prompts, not calibrated psychological scales.

For a prospective learner: push might be repeated failure to evaluate an AI feature; pull might be a rigorous studio with expert critique; anxiety might be the new school's lack of history or inability to complete the workload; habit might be relying on documentation and weekend experimentation. A sale becomes plausible when the forces favoring change outweigh the forces preventing it for that buyer in that occasion.

A founder who only increases pull through promotional promises but ignores anxiety and habit can achieve impressive interest and little purchasing. Better evidence, a smaller complete pilot, employer sponsorship, or different timing may resolve the actual barrier.

17. Circumstances often segment better than identity

The same person can have different jobs on different days. Someone who casually watches introductory lectures today may need expert architectural review after accepting a new responsibility. A young parent might buy convenience, taste, health reassurance, and affordability at different meal occasions. A firm may purchase broad employee awareness training and separately contract a high-stakes security capability assessment.

Segmenting by circumstances can reveal demand that demographic averages conceal. But not every situational segment is economically reachable or stable. Good segmentation also needs identifiable triggers, sufficient volume, distinct buying behavior, and an affordable route to the buyer.

18. The switching story is more revealing than the wish list

Reconstruct the sequence leading to a real change: first discomfort; event that made the situation unacceptable; passive search; active search; first discussion; shortlist; anxieties; final trigger; purchase; onboarding; use; post-purchase judgment. For people who did not switch, reconstruct why they stopped.

In the story, listen for **struggle moments**: "I had tried three tutorials, then the system failed in our release demo." "Our dean wanted the elective, but the faculty workload was impossible." "We tested two vendors, but legal could not approve either." These moments reveal causal bottlenecks better than generic answers about product preferences.

The goal is not to coax a dramatic story from every customer. Sometimes purchases are routine and minimally considered. The method is especially informative for consequential changes and product adoption.

19. Nonconsumption can be an enormous competitor

A customer who never buys has not necessarily chosen a named rival. They may be satisfied enough, unable to pay, unclear about the need, unable to find an acceptable offering, or blocked by authority. **Nonconsumption is a set of mechanisms, not one market segment.**

Distinguish constrained nonconsumers who *want* a solution but cannot access one from unconcerned nonconsumers who do not value the problem enough. The former may be served through lower-friction offerings, subsidies, employer-funded models, or new delivery. The latter may not constitute a near-term market.

Interview nonbuyers and people who abandoned transactions. Their stories challenge the false assumption that everybody who has a problem is already a customer in waiting.

20. A customer’s desired outcome is not their proposed feature

Imagine five learners request recorded lectures. One needs schedule flexibility. Another fears missing live instruction. A third wants to rewatch difficult derivations. A fourth lives with unstable connectivity. A fifth simply dislikes being questioned in class. One feature may imperfectly satisfy different underlying jobs.

A founder who categorizes all five as “wants recordings” misses the design choices. Some need asynchronous mastery checks; some need downloadable materials; some need a safer discussion environment. Observe **what breaks without the feature** and what alternative remedy would be accepted.

Translate feature requests into desired outcome statements, then design competing methods. This helps avoid building an omnibus product that satisfies the vocabulary of requests but not the causal needs.

21. The boundary of JTBD: it does not replace economics, power, or institutions

Jobs theory improves interpretation of behavior but does not tell us automatically whether a person can pay, whether a law permits the transaction, whether an administrator will approve, whether a trusted incumbent has distribution dominance, or whether the business can serve profitably. Two people can have the same job and opposite purchasing capacity. Three employees can have the same problem, but only one can control the training budget.

Do not turn every decision into an emotionally compelling narrative. A complete customer model combines jobs with budgets, process, observed alternatives, price, credible proof, access, and the surrounding market structure. The model becomes useful only when those components predict action.

Part III — Discovering Problems That Are Worth Solving

22. There is a hierarchy of problem evidence

A founder hears “I struggle with AI” and concludes demand exists. But a complaint does not establish urgency or an economic buyer. Separate **awareness**, **annoyance**, **costly inconvenience**, **active attempted remedy**, **budgeted priority**, and **actual purchase or substantial commitment**. Stronger levels generally indicate more actionable demand, though purchases can also reflect hype or mistaken beliefs.

A useful evidence ladder is: (1) abstract opinion; (2) specific recent incident; (3) measurable cost or consequence; (4) repeated workaround; (5) credible search or approval activity; (6) real expenditure; (7) repeated paid behavior under realistic alternatives. The ladder is a reasoning aid, not a fixed validation score.

For a new founder, recurring costly behavior is often more informative than a thousand survey endorsements. Nevertheless, a single large corporate purchase may be idiosyncratic; repeatability remains a separate question.

23. The anatomy of acute demand

An acute problem usually has some combination of **severity**, **frequency**, **time sensitivity**, **ownership**, **recognized consequence**, **insufficient alternatives**, **accessible budget**, and **the buyer’s ability to act**. It need not score high on every factor. Rare catastrophic incidents can be urgent; small recurring irritations can support large businesses when frequent and widespread.

Consider three software-learning situations. A hobbyist wants to understand AI better someday. A working developer repeatedly loses a weekend to model failures. A technical lead must demonstrate an auditable system in forty-five days or risk losing an important customer. These differ in urgency, context, perceived downside, and probable budget. The final one may be much more acute, but the buyer could prefer a consulting review to a six-month course.

Always add the crucial question: **acute for whom, and through which buying mechanism?** A high-stakes user problem without an authorized buyer is a valuable social problem and potentially a weak commercial entry point.

24. A problem’s cost is multidimensional

Costs can include money lost, time consumed, failures, safety exposure, delayed revenue, career risk, mental effort, shame, foregone learning, idle capacity, regulatory penalties, and strategic opportunity. You should record these separately before attempting to monetize them.

Ask customers for ranges grounded in actual cases: How many hours were spent last week? What work was repeated? What happened when delivery was late? Which invoice, deadline, or error report illustrates the

consequence? The founder must avoid converting every inconvenience into a dramatic salary-equivalent estimate; not every saved hour yields cash savings.

For education, loss of personal time and confidence can be consequential without being a direct money loss. Separate **personal perceived value**, **organizational cash economics**, and **societal benefit**. They can point in different directions.

25. Frequency, severity, and recoverability

A problem that happens daily but is minor may justify a low-priced habitual product. An annual audit failure can justify significant preventive spending. A one-time career transition can generate a major educational purchase. The economics depend on the event's severity, probability, reversibility, timing, and who bears the loss.

A proposed *pain score* that multiplies severity by frequency may be convenient, but it can be misleading because those scales are subjective and often non-linear. For initial discovery, write a short evidence-backed paragraph instead: what happens, how often, what it costs, what is already done, and why resolution matters *now*.

26. Workarounds are evidence of costly demand

A workaround is what customers do when no formal product fits, or when available products are unattractive. Examples: maintaining fragile Excel workbooks; hiring freelancers repeatedly; asking colleagues for informal reviews; manually reconciling AI outputs; assembling self-study playlists; faculty teaching a new elective outside their specialization.

The founder should observe **what the workaround actually costs and what it protects**. A spreadsheet may be frustrating yet flexible and trusted. A manual step may be legally required. A colleague's informal mentorship may create social value beyond technical advice. Replacing the workaround is only attractive if the new product preserves the desirable properties while improving the costly ones.

27. The cost of delay is a diagnostic, not a fear tactic

Ask: "What happens if this problem remains unsolved for thirty days, six months, or a year?" If the answer is "almost nothing," willingness to act soon may be low even when people admire a solution. If an expensive release is blocked, waiting may be unacceptable.

But beware of a founder who invents catastrophic scenarios that the customer does not believe or which the evidence does not support. The aim is to understand *existing consequences*, not intensify anxiety. Good early opportunities often have a naturally occurring clock: contract renewal, exam date, promotion cycle, procurement window, product release, business season, or staffing transition.

28. Budget is evidence of priority but not proof of fit

Ask not merely whether the company has money, but **which budget line already funds an alternative**. A firm might spend on contractors, training, cloud services, overtime, recruitment, rework, or consultant audits. A new offering may replace one of these expenditures or require creation of an entirely new category.

An existing budget can ease purchase because the economic buyer already understands the need. But incumbent commitments, approved-vendor rules, and purchasing cycles may make transfer difficult. A new category may be compelling and still fail because nobody owns its budget. Interview the payer and the approver before treating user enthusiasm as revenue.

29. Problem discovery is partly the study of failure

Success stories tell you what works in favorable cases; failures reveal where an activity breaks under constraints. Interview people whose projects failed, purchases were refunded, vendors were rejected, courses were abandoned, or initiatives never moved beyond a proposal.

Ask about **the sequence before failure**. Was the initial problem wrong, the solution untrusted, implementation too difficult, delivery inadequate, internal politics unresolved, or the job itself unimportant? Do not assume every failure proves a market gap; some show that serving the segment is too expensive or that adoption incentives are irreconcilable.

30. Distinguish a badly solved problem from an unsolvable business model

Customers can have genuine pain that a particular founder cannot serve profitably. A free educational intervention may create tremendous benefit but have no direct payer. Some enterprise problems require approvals beyond a startup's capability. Certain safety-critical services require credentials or infrastructure that cannot be improvised.

The entrepreneurial problem is to find an intersection of **consequential need, feasible remedy, reachable authority, credible evidence, and attractive delivery economics**. Do not confuse personal empathy with a mandate to accept an impossible contract. A founder can care about a problem and still conclude that another organizational form or funding source is needed.

Part IV — Customer Segmentation That Changes Business Decisions

31. Segmentation is a hypothesis about differences in behavior

The purpose of segmentation is not to create attractive personas. It is to group prospective customers whose **problems, choices, willingness to sacrifice, buying processes, or success requirements differ enough that the business should treat them differently**. Yankelovich and Meer argued for segmentation tied to purchasing behavior and decisions rather than decorative demographic descriptions [R8].

A segment is useful if understanding it changes at least one meaningful decision: product scope, quality standards, price, delivery, acquisition channel, sales process, support, or whom not to serve. “Ambitious millennials” may make a marketing presentation feel focused without altering delivery. “Employed backend engineers assigned their first AI production release, with no internal reviewer and under eight weeks of lead time” may imply a tightly scoped review-intensive product.

32. Seven lenses for dividing a market

Segmentation lens	What varies	What the founder might change
Demographic / firmographic	Age, profession, industry, size, region	Language, eligibility, legal compliance, access route
Situational / trigger	New job, product deadline, audit, childbirth, relocation	Buying timing, urgency, offer duration
Job / desired outcome	Exploration, skill proof, cost saving, risk avoidance	Product, assessment, success definition
Readiness / capability	Experience, prerequisites, maturity, resources	Onboarding, class level, technical scope
Behavioral / alternative	Workarounds, spending, tools, switching habits	Competitive positioning, migration, evidence
Buying structure	Self-payer, sponsor, committee, procurement	Pricing, contracting, sales, proof
Risk / trust requirements	Reversibility, status, criticality, hidden quality	Trials, guarantees, audits, references

These lenses can be layered but not infinitely multiplied. A segment defined by ten criteria may have only five people. Keep a meaningful distinction between **who could benefit**, **who is reachable**, and **who will actually buy**.

33. A segment must be identifiable and actionable

A founder can observe an interesting difference that no channel can target. For example, "engineers who privately feel inadequate" is psychologically plausible but difficult to identify without intrusive assumptions and may invite exploitative messaging. "Developers newly responsible for evaluating AI releases" may be accessible through role changes, relevant communities, approved professional networks, or self-selection through technical content.

A useful segment satisfies at least five tests: distinguishable decision behavior; a consequential common job; identifiable buying triggers; ability to reach with lawful, affordable means; and enough purchasing capacity to sustain the proposed model. Add an honest **delivery homogeneity** test: can one product serve the segment without unacceptable customization and exclusion?

34. Behavioral segmentation is stronger than arbitrary labels when it predicts action

Suppose an institute labels its potential customers beginners, intermediate, and advanced. Those groups may be pedagogically useful but insufficient commercially. An advanced engineer exploring theoretical statistics may not pay for a supervised AI project; an intermediate engineer with an immediate deployment responsibility may.

A better working cross-tab combines **capability readiness × urgency of job × payer access**. High readiness, high urgency, and a clear payer may indicate a plausible paid specialist studio. Low readiness, high urgency, and limited time may indicate a customer whose problem is real but whose desired transformation is infeasible in the proposed format. High readiness and low urgency may fit free public material or a low-touch product.

Use the cross-tab to identify *testable clusters*, not to label individuals permanently.

35. Personas are useful as hypotheses, dangerous as evidence

An invented persona can help a team remember a coherent customer situation. But invented names, salaries, preferences, and inner motives can acquire false authority through repetition. Write personas as **evidence-backed decision profiles**: observed trigger, job, current alternative, sacrifices, purchasing authority, important unknowns, and documented variation.

Separate quotations from interpretation. A research participant's phrase "I want mentorship" is evidence that they used those words, not proof that mentoring is the correct intervention. A team can use fictional composites for communication only when they are labeled as composites and their claims trace back to actual interviews or behavior.

36. Segmentation is dynamic and must be rechecked after purchase

The people who click an advertisement, complete an eligibility form, pay full price, attend every class, complete assessed work, and refer others may be different subsets. If only the enthusiastic but unprepared enroll, you have a selection problem. If strong learners enroll but cannot protect time, you may have a scheduling or commitment problem. If the highest-converting leads demand placement guarantees, you may be attracting the wrong job.

Do not conclude a segment is attractive merely because it responds to marketing. Check **acquisition, conversion, delivery, realized value, cost to serve, and post-purchase regret**. Segmentation should be updated when evidence contradicts the initial definitions.

37. Multiple-sided markets require separate segmentations

A marketplace for experts and learners has two distinct sides with different jobs and incentives. An enterprise training provider may have an economic buyer, a technical manager, and individual employees. A university service provider may sell to administrators and teach students. Treating all as one customer group conceals conflicts.

Make a separate value, proof, and veto map for each stakeholder group. Then study the **joint feasibility** of their incentives. A curriculum can delight students and burden faculty; a manager can want rapid training and employees can lack time; an employer can benefit from capability gains that individuals cannot afford to finance. In multi-sided settings, the product often must solve a coordination problem as well as an individual job.

38. Segmentation can be a strategic refusal

A founder who serves everyone is often forced to compromise the very service quality that could make the business distinctive. An advanced AI systems course cannot efficiently accommodate someone who has never programmed if it promises deep technical work on a fixed schedule. A boutique consultancy cannot profitably accept every adjacent request if each destroys reusable specialization.

Explicitly define **not yet served**, **not suitable for this offer**, and **never ethically appropriate**. For education, rejecting someone from a demanding cohort should mean the course cannot responsibly serve them as configured, not that they lack potential. When viable, offer a bridge or an alternative. The goal is coherent value creation, not theatrical exclusivity.

Part V — Discovering Customers in the Field

39. Interviews are instruments for investigating reality

Steve Blank's customer-development approach emphasizes leaving internal planning behind and testing hypotheses with customers and other business-model stakeholders [R9]. The key word is **testing**. An interview should reduce uncertainty about a consequential decision, not serve as a performance in which the founder seeks permission to be excited.

Before interviewing, write your main assumptions: who has the problem; what they currently do; how often it occurs; what it costs; who pays; what triggered consideration; why they might switch; why they might refuse. Then decide what experience would contradict each assumption. Without prior hypotheses, founders easily mistake an engaging conversation for useful discovery.

40. Four fundamentally different research conversations

Exploratory interview: Learn how a customer's world works, what difficult episodes recur, and which alternatives exist. Minimize pitching. **Buying-decision reconstruction:** Study a recent actual choice, including rejected options and constraints. **Workflow or contextual interview:** Observe the real task and investigate discrepancies between reported and actual behavior. **Solution-evaluation interview:** Present a specified prototype or offer and investigate trade-offs, skepticism, and practical adoption.

Each produces different evidence. Do not present a polished pitch and then call positive reactions unbiased problem discovery. Do not use a five-person usability test to claim five percent conversion. Name the research method in the notes so subsequent readers know what was learned.

41. Who should you interview first?

A first-time founder may have no customer list. Begin by mapping **situations that recently occurred**, not simply profiles that sound suitable. For an AI education hypothesis, identify people who recently attempted to deploy AI features, sought external learning, paid for courses, declined courses, or received employer sponsorship. For colleges, seek academic coordinators who recently approved or rejected external delivery. For enterprise buyers, find managers who recently purchased or canceled external training or pilots.

Include five useful types: **recent buyers, recent nonbuyers, dissatisfied incumbents, active workarounds, and relevant decision-makers**. Do not recruit only enthusiastic followers or personal friends. Their views may be convenient and atypical. Seek contrasts: people who succeeded without your product, people who experienced the problem but did nothing, and people who wanted help but could not get approval.

42. Recruiting when you begin with no audience

Possible sources include permission-based professional connections, community administrators who approve research invitations, university departments, relevant professional associations, participants at technical events, referrals to people with a *specific recent experience*, and interviews with knowledgeable suppliers or intermediaries.

Write a recruitment request that states **who you want to learn from, what recent situation qualifies them, how long the conversation takes, how notes are handled, and whether the interaction is research or selling**. Do not imply a partnership or institutional affiliation that does not exist. Do not treat old prospect databases as permission to contact people; verify lawful and appropriate use.

A founder may offer reasonable research compensation or a useful report, transparently disclosed. This can improve participation but might alter who responds; note the selection mechanism.

43. Sampling is a design problem, not a magic interview count

Ten interviews in a narrow group may reveal a striking failure mode. Ten interviews cannot estimate population willingness to pay accurately. Twenty friends who say yes may be less informative than five recent purchasers and five recent rejecters. There is no universal sample size for customer discovery. The required number depends on heterogeneity, stakes, purpose, information richness, and whether new interviews change the conclusions.

A useful iterative approach: start with a few deliberately contrasting participants; examine emerging patterns; recruit people who could refute them; continue until the decisions that matter are supported by multiple independent observations, then test behavior. **Qualitative pattern recognition is not statistical representativeness**. Large-scale sizing, conversion rates, and segment prevalence require appropriately designed quantitative evidence. Guidance about five-person usability testing is specifically about finding usability problems in repeated small studies, not proving startup demand [R10].

44. The interviewer’s job is to reconstruct, not persuade

The most dangerous leading question is: “Would you pay for our excellent mentor-led AI course?” It embeds approval of the product in the question and makes the interviewer an interested salesperson. Replace it with: “Tell me about the last time you tried to learn something difficult for work. What happened before you decided to seek help?”

Use concrete past events, actions, and comparisons. Ask *what* happened, *who* was involved, *how* the person decided, and *what* they did next. “Why?” can be useful but may invite retrospective rationalization; pair it with timeline and artifact questions. Do not turn uncertainty into a courtroom interrogation. A relaxed, respectful conversation improves both ethics and data quality.

45. The thirty-to-forty-five-minute discovery interview

Opening (3–5 minutes). State the research purpose, confidentiality boundaries, whether recording is requested, and that there is no need to praise the founder's idea. Ask permission before recording. **Recent concrete event (10 minutes).** Ask for the last occasion when the problem occurred; establish trigger, setting, stakes, people, and constraints. **Current remedy (10 minutes).** Reconstruct exactly what they did, used, paid, waited for, or abandoned. **Decision journey (10 minutes).** Understand search, shortlist, trusted advice, risk, vetoes, and reasons for action or inaction. **Consequences (5 minutes).** Explore what changed and what remains unsolved. **Close (3–5 minutes).** Ask what important question was missing and request an introduction to a contrasting participant only with appropriate consent.

This is a flexible guide, not a script to read mechanically. Follow surprising facts. Avoid showing the solution until the participant has described the unprompted problem; if you also need to test a concept, clearly mark the transition to a different research phase.

46. Interview questions that reveal real decisions

Useful prompts include: "What prompted you to look *then*?" "What were you using before?" "Show me what happened, if it is safe to share." "How much time did that take?" "What did you try after it failed?" "Who else was involved?" "What alternatives made the final shortlist?" "What nearly prevented the decision?" "What was the exact condition required for approval?" "Did you spend money or obtain a budget?" "If you did nothing, what happened?" "Knowing the result, would you make the same choice?"

Avoid asking everything. One well-investigated episode can be more useful than twenty opinions. Record exact language sparingly and accurately, distinguish memory from contemporary artifacts, and ask permission before recording sensitive professional details.

47. The nonbuyer interview may be more valuable than the buyer interview

Buyers can explain why the product felt attractive. Nonbuyers identify the boundary of the market. Ask someone who investigated a category but declined: What initiated the search? Which options were credible? At what point did the option become unacceptable? What did they do instead? Did the problem disappear, persist, or worsen?

For a failed college proposal, the answer might be "we liked the curriculum but cannot approve a vendor this semester," "faculty would refuse an external instructor," "students cannot absorb the additional workload," or "the department has no budget." Those mechanisms imply different opportunities. "Not interested" is often too coarse to guide a decision; but do not assume every polite decline conceals a solvable objection.

48. The lost-deal interview tests the salesperson's story

If a prospect says "too expensive," sales teams may code "price objection." A neutral interviewer may discover the customer doubted the provider's operational capacity, disliked an inflexible schedule, or never received procurement approval. Reconstruct the decision without pressuring the person to reopen it.

Compare the customer's account with contemporaneous events: emails, proposal dates, evaluation criteria, and steps actually completed where access is legitimate. Do not treat every stated explanation as literal truth or every lost deal as preventable. Sometimes the competitor was genuinely better, and that is important information.

49. A bad interview and a better interview

Bad version: Founder: "Would you like personalized, industry-grade mentorship in an AI course?" Prospect: "Yes, definitely." Founder: "Would ₹45,000 be reasonable if it changed your career?" Prospect: "Maybe, if it worked." Founder: "Great, another validated lead."

The conversation established little. It presented a flattering hypothetical product, an extreme possible benefit, and no actual choice.

Better version: Founder: "When was the last time you paid for technical training?" Prospect: "Last year, a self-paced cloud course." Founder: "What led to it?" Prospect: "My team migrated a service." Founder: "What happened after the course?" Prospect: "I watched half, then asked a senior colleague when things broke." Founder: "What did that colleague help with?" Prospect: "Deployment reviews and IAM mistakes." Founder: "Would it be possible to describe one non-confidential example? Who approved the time and budget?"

Now a testable hypothesis appears: **practical review at the point of implementation may be more valuable to some learners than additional content**. But one story does not establish willingness to pay or the best product. Follow it with contrasting cases and a realistic offer.

50. What not to do in interviews

Do not ask respondents to design your business, compliment your mission, predict behavior years ahead, or name a price for an undefined offer. Do not ask manipulative questions, request confidential artifacts without authorization, record secretly, make it seem that research participation obligates a purchase, or use vulnerability to create artificial urgency. Do not interpret silence as agreement.

Avoid **solution contamination**: once the participant hears "our course has weekly experts and hands-on labs," they may use your vocabulary when describing their problem. Separate unprompted statements from reactions after exposure. A buyer's ability to repeat your pitch is not proof the original job existed.

51. Observation can correct confident but inaccurate accounts

People may not know their own workflows perfectly. They may estimate time poorly, forget hidden assistance, overlook informal steps, or describe official procedures that differ from reality. **Contextual inquiry** combines observation and questions while users perform actual tasks [R11]. For a developer, watching the steps leading from a failing AI evaluation to an attempted fix may reveal tools, interruptions, approvals, and tacit expertise missing from interviews.

For an education provider, observing a learner attempt a realistic task is extremely informative. You may discover that the purported knowledge gap is actually poor feedback, insufficient practice, incorrect tool use, or organizational restrictions. When observation touches employer code, customer data, or student work, use authorized environments, anonymized examples, or synthetic tasks; no entrepreneurial insight justifies violating confidentiality.

52. The artifact method: what customers leave behind

Documents, logs, invoices, screenshots, plans, calendars, purchase orders, error reports, support tickets, decision memos, proposals, abandoned applications, and submitted assignments can reveal the actual shape of a problem. An artifact can corroborate a memory, show time and money already invested, and expose a constraint that an interview misses.

Artifacts are **contextual evidence**, not permission to collect personal information indiscriminately. Ask for the minimum necessary, redact sensitive details, obtain consent and authority, store securely, and establish retention. A non-confidential workflow diagram may convey more useful insight than a full repository you should never have received.

53. Diary studies and longitudinal observation

For intermittent or habitual problems, one retrospective interview can miss patterns. Ask a small group to log events as they occur: trigger, context, tool used, attempted remedy, elapsed time, frustration, outcome, and whether a purchase was considered. A week of recorded study interruptions may show why a working professional cannot complete an online course even though they sincerely intend to.

Diaries have compliance burden and reactive effects: observing behavior can change behavior. Keep logs short, consent clear, and interpretation modest. Use them where repeated occasions matter, not because more data always means better insight.

54. Search, community, and public conversation as secondary evidence

Search queries, public forum questions, reviews, support communities, and competitor complaints can generate hypotheses about recurring jobs and language. They are attractive because they are accessible before the founder has customers. But selection is severe: loud complainers differ from quiet satisfied users; algorithmically amplified posts are not representative; a viral complaint may be rare.

Use secondary evidence to recruit relevant interviews and identify actual alternatives. Check whether people pay, change tools, or invest effort to address the problem. Public data can indicate **what is discussed**; it usually cannot alone indicate **who is ready to buy**.

55. Customer support is continuing research, not merely service recovery

After launch, the most honest customer information often appears in support requests, refund explanations, complaints, missed deadlines, and product workarounds. These reveal gaps between intended and actual use. A cancellation that says “no time” may represent poor scheduling, weak urgency, mis-set expectations, or a life event. Examine the incident instead of mechanically assigning blame.

Create a taxonomy of observed failure modes, linked to specific customer contexts. Feed the evidence into product, admissions, marketing, and operational design. Avoid punishing employees for surfacing problems; otherwise the organization learns to hide negative evidence.

56. Field notes should separate facts from interpretation

A useful interview record has four layers: **verbatim or close paraphrase of what was said**, **observed actions/artifacts**, **researcher’s interpretation**, and **unresolved alternatives**. For example: “Respondent abandoned two online courses; reports needing code review; spent four evenings on an evaluation bug” is not identical to “will pay ₹50,000 for mentorship.”

Immediately after an interview, write the event timeline, actual alternatives, exact purchase or nonpurchase, roles, consequences, contradictions, and next question. Record source type and date. Distinguish the customer’s belief that a course improved their career from independently demonstrated causal improvement.

57. A better synthesis process than counting mentions

Teams often tally feature mentions: eight asked for mentors, six for projects, four for certificates. This counts words rather than decisions. Eight participants might mean different things by “mentor.” Instead, compare **episodes of struggle**, their contexts, the costs incurred, the chosen remedy, and the reasons it failed or succeeded.

Group cases by the mechanism that explains action. For example: “cannot self-diagnose architecture errors under time pressure” differs from “needs credential recognized by an employer,” even if both participants request mentorship. Build a matrix with cases in rows and hypothesized causal variables in columns. Seek contradictions deliberately; do not create a theme only from memorable quotes.

58. Saturation is local and conditional

When the same problem appears in repeated interviews, the founder may be tempted to declare discovery complete. But saturation depends on **which population, which context, which question, and which level**

of detail. Ten interviews among working developers do not establish the motivations of college students, corporate sponsors, or learners outside the recruitment channel.

A responsible stopping rule asks: Are the important distinctions becoming stable within this narrow group? Have we investigated contradictory cases? Can we predict the next person's existing alternative and likely constraints? Do we have enough evidence to justify a cheap behavioral test? That is more meaningful than an arbitrary interview quota.

59. Surveys answer different questions from interviews

Interviews are useful for discovering mechanisms, vocabulary, and unexpected alternatives. Well-designed surveys can estimate frequencies and differences in a defined population if sampling, question wording, nonresponse, and measurement are appropriate. A survey sent to self-selected followers cannot produce representative market incidence merely because a thousand people respond.

Survey questions should be piloted for comprehension and bias. The US Census Bureau's questionnaire evaluation guidance emphasizes cognitive interviews, pretesting, question-order effects, and other sources of measurement error [R12]. Use clear behavioral recall windows ("during the past thirty days") instead of vague "often"; distinguish "considered buying" from "purchased"; and avoid combining two questions into one.

60. Design a survey around decisions rather than applause

A poor survey asks respondents to rate how revolutionary an AI program sounds. A useful one might ask: In the last twelve months, which learning or implementation problems occurred? Which remedies were actually used? What money was spent? Who paid? What prevented purchase? How many hours were committed? Which options were considered? What is the respondent's role and recent decision context?

When asking about a proposed solution, define the offer precisely, include realistic alternatives, and state the hypothetical character of the response. Responses may reveal patterns, but do not count stated purchase intention as realized revenue. More elaborate stated-preference methods may help with larger, well-sampled product studies; they cannot rescue an undefined offer or an unrepresentative sample.

Part VI — Revealed Preference, Commitment, and the Limits of Self-Report

61. What people say is evidence; what they do is different evidence

Suppose an interviewee says they would pay ₹60,000 for a serious course. That response is information about their current imagined value under the interview conditions. It is **not** evidence of a completed ₹60,000 purchase. The hypothetical course may have vague faculty quality, uncertain timing, undefined workload, and

an unknown competitor set. People may wish to appear enthusiastic, overestimate future discipline, or fail to imagine the real cost of spending money.

Economic research has found hypothetical bias in various stated-preference settings [R13], but there is no universal factor by which you should discount survey answers for all educational purchases. The practical conclusion is more modest: **observe real choices where feasible**, and classify evidence by how closely it matches the actual decision you need to predict.

62. Revealed behavior is more grounded, but not infallible

Paying for a product is stronger evidence of willingness to pay than saying one might. Yet it can also be distorted. Someone buys an inferior product because of misinformation, pressure, lack of alternatives, a temporary discount, an employer's budget, or a friend's request. A person who pays once may regret the purchase. A company signing a contract may never deploy the tool.

Thus examine the entire trajectory: **who paid, what they believed, what alternative they rejected, what they used, what value they obtained, whether they would repeat, and what it cost to acquire and serve them**. Revealed preference helps identify actual trade-offs; it does not prove that customers are well served or that demand will repeat at scale.

63. A hierarchy of behavioral commitments

Observable action	What it can support	What it does not establish
Compliment or social-media like	Topic salience, social engagement	Purchase intent
Reading a technical article	Interest in content or problem	Desire for a paid service
Completing a substantial diagnostic	Willingness to devote time	Willingness to pay a larger fee
Requesting a proposal	Some consideration, depending on context	Procurement authority or budget
Seeking employer approval	A possible organizational champion	Approval will be granted
Refundable deposit	Limited financial commitment	Full-price sustained demand
Paid pilot at disclosed price	Real exchange under stated conditions	Repeatable growth or long-term satisfaction

Observable action	What it can support	What it does not establish
Full-price purchase and participation	Real acquisition and initial use	Durable outcome or referral
Renewal, reorder, voluntary reference	Observed repeated value and/or trust	Causal advantage or universal market fit

These are not exact rungs that every customer must climb. An enterprise pilot may be meaningful even when a free event is irrelevant. A consumer purchase may occur in seconds. The ladder's use is to prevent a founder from treating one type of action as proof of another.

64. Past behavior, present constraints, and future hypothetical choice

Past purchases reveal what a buyer was willing and able to trade in an earlier context. They may not predict a new purchase after technology, income, employer policy, or needs change. Present constraints reveal what is possible now. Hypothetical future choices reveal reactions to a proposed configuration. All can be useful if kept distinct.

A well-designed interview asks three separate questions: **What did you do last time? What is materially different today? What would you do under this precisely stated new offer?** Do not collapse them into "the customer wants it." The first is historical behavior; the second changes the conditions; the third is a forecast awaiting a genuine transaction.

65. Money, time, and authority are different currencies

A person may commit two hours to a free workshop yet refuse a ₹5,000 course. Another may gladly pay ₹25,000 but never complete an entry assignment. A college dean may promise a meeting but have no authority to sign. These commitments are not equivalent.

Whenever a prospect takes action, record **which scarce resource they surrendered**. Was it attention, a calendar slot, professional reputation, budget approval, cash, access to data, recurring behavior, or a contractual obligation? The closer the sacrifice resembles the one your eventual business requires, the more diagnostic it is. A person signing up for a webinar has not committed to six months of demanding independent work.

66. A real experiment must preserve the important conditions of purchase

Suppose you want to test a ₹50,000 program. A free two-hour workshop may validate instructor competence and problem relevance, but not the long-course commitment. An extremely discounted founding offer may prove some demand, but customer mix and expectations may change at full price.

Define the intended buyer, real deliverables, required work, instructor commitment, schedule, assessment, refund policy, and price. Ask for a genuine payment only when you can deliver responsibly. Compare responses across realistic alternatives. Small sample sizes and founding novelty limit inference, but **one well-defined paid transaction can establish a more meaningful fact than hundreds of vague endorsements**.

67. Why “would you pay?” is a poor primary pricing question

People may respond with a socially acceptable number, anchor to the founder’s suggestion, or imagine a different product. Instead ask: “What did you last purchase to solve this problem? What was included, and what did it actually cost?” “What other methods did you consider?” “What would the decision depend on?” Then test a precise paid offer or a transparent scope-and-price comparison.

When price research becomes more sophisticated, Chapter 15 will examine willingness to pay in depth. Here the principle is to preserve the real decision structure. **Price is a trade-off against a specific alternative and total burden**, not a personal preference detached from time, credibility, and outcomes.

68. Why conversions can lie

Imagine 100 people arrive at a course page and eight buy. Eight percent looks attractive until you learn that seven buyers are personal friends, six paid half price, and four cannot meet the prerequisites. The conversion rate is a property of a specific funnel with a specific population and offer, not a universal forecast.

Record denominator quality: eligible prospects, acquisition source, prior relationship, stated job, budget, readiness, paid amount, cancellation, support load, and verified learning. Compare similar groups and avoid extrapolating warm network conversion to cold audiences. For B2B, record whether proposals were requested by genuine budget owners or by curiosity-driven intermediaries.

69. Adoption, activation, and realization are separate choices

A customer may buy, fail to use, and obtain no value. Software users may sign up without completing a workflow. Learners may enroll but not complete assignments. Enterprises may sign training contracts but not release staff from work. A founder who measures only payment misses whether the customer’s intended job is ever performed.

A complete customer model therefore predicts **adoption → first valuable use → repeated engagement → delivered outcome → continued choice or recommendation**. The causal requirements vary. The school needs learner time and instruction; software needs usable onboarding and integration; FMCG needs product satisfaction and availability. The earliest real customer research continues after payment.

70. Design experiments to disprove your favorite story

If the hypothesis is that developers are desperate for expert architecture feedback, offer an actual paid review of a bounded non-confidential problem and watch whether qualified developers schedule and pay. Compare

with a competing offer centered on structured independent practice. If they admire the concept but do not buy either, investigate urgency and the trusted free alternative.

A strong experiment specifies **belief, buyer, offer, observed action, time and money limit, threshold for changing the plan, and alternative explanations**. At small samples, thresholds are operational safeguards, not statistically estimated universal rates. Record what would make the hypothesis false before you collect responses.

Part VII — Alternatives, Choice Architecture, and Purchasing Systems

71. Competitors are the solutions customers can actually choose

A founder researching online education might list well-known training companies. A real buyer's consideration set might include YouTube, an internal colleague, a paid mentor, a university program, an open-source contribution, a project at work, doing nothing, and postponing learning. These alternatives compete for scarce money and time even if their business models differ.

The correct competitor question is **"What did this customer use or seriously consider instead?"**, not "Which other company sells our category?" Demand is defined through substitution in a specific decision. The same product may have different competitor sets for different jobs.

72. Separate nominal competitors from effective substitutes

A nominal competitor has a similar product description. An effective substitute can accomplish the job well enough under the buyer's constraints. A self-paced video may be a weak substitute for individualized expert review, yet a sufficient substitute for learning basic syntax. A consultant can substitute for a software tool in a short project, but may be uneconomical for ongoing high-volume processing.

Map substitutes by **job coverage, performance, time, risk, total cost, familiarity, and accessibility**. A startup can win without being globally superior if it is better along the dimensions that matter in one buying occasion. It can also lose to a technically inferior alternative that is easy to trust and deploy.

73. The "last satisfactory alternative" test

Ask the customer to identify the option they would choose if yours vanished tomorrow. Then ask: what does that alternative do satisfactorily? Where does it fail? Why has it remained acceptable? How would the customer know your option did better? What switching cost remains?

This prevents a common delusion: describing everything wrong with the incumbent while ignoring the reasons it survives. A clumsy spreadsheet may be transparent, flexible, familiar, and already approved. A

lecture-based college program may be financially predictable and easy to timetable. To displace them, the new provider must preserve enough value while changing what matters.

74. The choice set is often constructed during the decision

Buyers rarely compare every possible product. They learn a category, discover a few options, seek advice, shortlist, and revisit criteria. What gets considered depends on distribution, terminology, trust, and timing. An unknown provider can be outstanding and absent from the choice set.

Ask “Which alternatives did you first know about?” separately from “Which reached the final shortlist?” and “Which did you reject before comparison?” This helps distinguish **lack of awareness, lack of perceived relevance, lack of credibility, and true competitive disadvantage**. Chapter 20 will treat distribution; here the founder’s task is to understand the customer’s real search process.

75. A purchase is sometimes a team sport

For an organizational purchase, no single person’s preference may be decisive. The user wants convenience; the manager wants output; security wants risk controls; legal wants defensible terms; procurement wants approved process; finance wants budget discipline; the executive sponsor wants accountability.

Map who **initiates, champions, evaluates, signs, pays, implements, and can veto**. A solution that meets one role’s needs and fails another’s is not ready for purchase. Ask separate actors rather than relying on a single champion’s report of what everyone else thinks.

76. The six organizational buying jobs

Gartner describes complex B2B purchasing as a non-linear set of jobs including **problem identification, solution exploration, requirements building, supplier selection, validation, and consensus creation** [R14]. A buyer may revisit these repeatedly. This is a practical corrective to the assumption that a committee moves smoothly from awareness to purchase.

For a startup, the implication is not to build a giant sales bureaucracy. It is to discover **which buying job is unfinished**. Perhaps the sponsor cannot quantify the problem; perhaps procurement demands a compliance document; perhaps faculty have not agreed on timetable changes; perhaps finance cannot see a budget. Providing the right proof at the right stage can be more valuable than another persuasive meeting.

77. The champion and the buyer are not interchangeable

A motivated engineer may champion AI training because they want technical growth. Their manager may favor a workshop because it solves a team delivery problem. A training budget owner may require multiple employee participation and measurable outcomes. A procurement officer may insist on an approved provider. The founder must speak to each in the language of their actual responsibility.

A genuine champion can help explain internal decisions and introduce stakeholders. But enthusiasm is not a signed contract. Ask whether the champion has previously secured similar purchases, who controls spend, what approval milestones exist, and what objection could still stop the agreement.

78. Purchasing authority can be distributed and conditional

Authority varies by amount, category, department, and fiscal calendar. An engineering manager may approve a small tool but not an external vendor with access to employee data. A dean may be able to propose a curricular partnership but need a committee's endorsement. A consumer learner may choose a course but rely on family funds.

Your interview should reconstruct **actual prior purchases at similar financial and operational stakes**. Who signed the last training contract? How long did it take? Which documents were mandatory? What derailed previous proposals? Abstract titles such as "Director" and "Head of Department" do not reliably reveal approval powers.

79. The decision journey is not always a funnel

Funnel stages are convenient for measurement, but real buyers may search, pause, consult, revisit, change requirements, and reenter later. A prospect can be convinced of value but blocked by timing; another can unexpectedly buy after a trusted referral. The founder needs a **decision process map** alongside CRM stages.

Record typical trigger, information sources, evidence needed, actor participation, budget release, alternatives, sticking points, and purchase timing. Use the map to decide which product or proof to test first. Do not assume the path is identical for all segments.

80. The last mile of buying matters

Even after customers decide "this looks useful," payment, contracts, refunds, scheduling, invoice formats, tax treatment, vendor registration, onboarding, security paperwork, or campus timetabling can stop them. Some founders call all these "administrative details" and then wonder why deals do not close.

For individual education, visible tuition, payment security, time zone, class schedule, and withdrawal terms matter. For colleges, faculty rosters, curriculum approval and academic calendars matter. For enterprise engagements, data handling, deliverables, liability, procurement and acceptance criteria matter. Buyer understanding includes how an intention becomes a legally and operationally feasible commitment.

81. The purchase can solve a political problem rather than a technical one

A team may buy outside consulting partly to secure an independent opinion that is easier for leadership to accept. A manager may choose a famous vendor because its recommendation is easier to defend. A university

department may favor an external partner to show industry relevance while preserving internal authority. These motives cannot be reduced to technical performance alone.

Rather than cynically exploiting politics, learn **whose incentives must align**. An offer may need clear ownership boundaries, faculty co-design, documented evaluation, or a pilot small enough that skeptics are willing to observe. If no legitimate arrangement resolves the conflict, the purchase may not be attainable.

82. Procurement creates a second product: the evidence package

For high-stakes buyers, the business sells both the underlying service and a credible, reviewable case for adopting it. This may include company identity, relevant experience, sample deliverables, security and privacy controls, instructor qualifications, workplan, price breakdown, reference permissions, legal terms, and acceptance criteria.

The point is not to imitate an enterprise vendor's 100-page packet prematurely. It is to identify the **minimum documentation for the actual decision**. A tiny technical pilot with no sensitive data may require far less than a deployment touching production systems. Customers often reveal the correct scope through the approval history of comparable purchases.

83. Different reasons to say no imply different business decisions

Customer says or does	Plausible cause	Investigation	Potential response
"I can learn this for free"	Substitute sufficient or paid value unclear	What did they accomplish alone?	Differentiate feedback, narrow job, or concede market
"No time"	Low priority or real schedule constraint	Actual calendar and prior commitments	Shorter complete offer or different segment
"Too costly"	Budget, weak value, uncertainty, alternative	Past spending, buyer, preferred alternative	Change payer, scope, proof or price hypothesis
"Send a proposal"	Genuine procurement or polite deferral	Who evaluates? When? What next action?	Obtain actual decision process
"Our faculty must approve"	Stakeholder veto	Which decision and evidence?	Co-design or defer launch
"Security will not permit it"	Governance constraint	Specific permitted data and system access	Redesign pilot; do not bypass controls

Customer says or does	Plausible cause	Investigation	Potential response
“We already have a vendor”	Satisfaction, switching cost or contract	Renewal date and unresolved pain	Find genuine underserved occasion
Paid but never activated	Onboarding, time, mis-sold product	Observe first use and commitments	Fix activation and qualification

None of these diagnoses should be assumed from one phrase. Treat them as hypotheses to investigate through the decision history and observed behavior.

Part VIII — How Customer Understanding Changes by Industry

84. Consumer staples: the decision may last seconds

A new snack or cleaning-product founder should observe **purchase occasion, shelf availability, pack size, price sensitivity, habit, sensory experience, and repeat purchase**. Asking shoppers for elaborate narratives about brand philosophy may explain little. A shopper can sincerely prefer one formulation and still buy another because it is in stock and familiar.

Research methods include in-store observation with appropriate permission, small product trials, receipt-based purchase history, retail sell-through, and repeat behavior after promotions end. Do not confuse a distributor accepting inventory with consumers purchasing it. For frequently bought goods, the most telling customer is often the **second buyer**, meaning the first purchaser returning without a launch incentive.

85. Luxury and identity goods: recognition is part of the value

Customers buying some luxury products care about provenance, craft, scarcity, experience, social recognition, gifting, and self-expression. An unknown premium brand cannot infer demand from respondents saying its design looks elegant. Observe actual purchases against known alternatives, willingness to wait, repeat patronage, gifting behavior, repair expectations, and reactions to provenance evidence.

These customers are not homogeneous. Some prioritize craftsmanship and private enjoyment; others buy a visible signal. A high-price label alone does not create a shared cultural meaning. Interview where the product is used and who, if anyone, sees it.

86. B2B software and AI: study the workflow, not the feature request

Developers may request more model options, buttons, and integrations. Observe what happens when a specific job fails: a release is delayed, a compliance check cannot be completed, a document cannot be reconciled, or a system lacks reliable evaluations. Identify handoffs and human workarounds. Software customers buy a reliable **workflow outcome**, sometimes embedded in a large technical environment.

Investigate deployment rights, security, integration effort, procurement, build-versus-buy, time to first value, monitoring, support, and cost after automation. A free tool may be more attractive than a paid product when enough functionality exists, even if the paid product's user interface is better. Study which limitations are genuinely costly.

87. Consulting and expert services: the client buys judgment under uncertainty

A boutique consulting client often needs a problem diagnosed or a consequential decision implemented. They may be unable to assess quality before engagement, so prior work, method transparency, references, contracts, and the professional risk of choosing the wrong adviser loom large.

Interview recent buyers about what triggered the engagement, what in-house alternatives failed, who sponsored the cost, whether the result was implemented, and how the client judged success. Look for a **bounded, independently useful diagnostic** that can provide immediate evidence without requiring an open-ended retainer. But avoid assuming that every client wants standardized consulting: some pay precisely for scarce tailored judgment.

88. Marketplaces: there are two customers and often three failures

Marketplaces must understand buyer demand, seller supply, and the trust/coordination mechanism between them. A buyer wants quality, availability, price, and recourse. A supplier wants credible demand, reliable payment, manageable fees, and fair rules. A beautiful platform can have many sign-ups and almost no completed transactions.

Study **search-to-match, matching quality, delivery, repeat, multihoming, and dispute resolution**. Concentrate initial research within one atomic market—one narrow geography, job, or category—where actual transactions can happen. Don't treat seller interest as proof of buyer demand or vice versa.

89. Healthcare, finance, and other high-stakes credence services

Users may struggle to evaluate quality even after receiving a service. Safety, professional standards, informed consent, regulation, liability, and independent evidence matter. A customer's anxiety is real, but monetizing fear through unverified guarantees is dangerous.

Discovery must involve legally authorized professionals and domain-specific research safeguards. Observe decision structure—patient versus provider versus insurer; depositor versus adviser versus regulator—and how

evidence is evaluated. A founder should never infer permission to offer regulated services from a generic finding that customers want them.

90. Local services and industrial goods: time and reliability can be the job

A neighborhood repair service may win because it arrives when needed, explains its price, and fixes the problem without repeat visits. An industrial components supplier may win because documentation, tolerances, delivery, and credit terms keep a production line operating. Both require different research from venture software.

Observe **arrival time, missed commitments, quality variance, travel costs, inventory failures, buyer credit cycles, and trust in local referrals**. The actual decision may be governed more by geographical access or procurement convenience than by the nominal object's features.

91. Education is both a transformation market and a trust market

Education customers seek future capability, confidence, opportunity, credentials, belonging, and sometimes relief from career uncertainty. Learning is co-produced: the learner's baseline, effort, support, environment, and labor market help determine results. Buyers often cannot verify the institution's causal contribution even after completing the course. The syllabus may be inspectable but transformation is uncertain.

A founder must uncover not just *why buyers want education* but **which outcome they will invest time to pursue, how they learn today, what they can actually sustain, which proof they require, and whether the institution can deliver that outcome with integrity**. A learner's expressed anxiety about AI is not proof that a six-month course is the right solution. A paid course is not automatically better than self-study or a workplace project.

92. When software replaces content, the customer's job may migrate

Generative AI can make explanations, code scaffolding, and generic practice more accessible. This can change what customers regard as scarce. Some may no longer pay for basic access to information; others may increasingly value robust assessment, accountability, authentic problems, advanced feedback, and trustworthy human judgment. These are hypotheses to test within each segment, not universal predictions.

Ask learners and employers **what they can already accomplish with AI, where AI assistance breaks down, and which human capabilities still require development**. Observe real tasks rather than rely on forecasts about "jobs of the future." As the underlying technology changes, customer understanding must be refreshed rather than preserved as a static persona.

Part IX — Bayes Institute: An Extended Customer-Understanding Laboratory

93. The honest starting position

Bayes Institute aspires to help people develop meaningful technical capability and eventually become a research-active STEM institution. Its founding mission is: **"Bayes Institute exists to create the conditions in which human possibility becomes human capability."** That mission supplies direction, but it is not customer evidence. The founder brings meaningful experience in software engineering, AI, statistics, teaching, and curriculum development. The institution itself is still early; it cannot presume established alumni outcomes, degree-granting authority, trusted employer recognition, repeatable customer acquisition, or paid partnerships that have not been signed.

The immediate operating constraints include limited time alongside a full-time professional role, an asset-light approach, careful cash use, no placement guarantees, and a preference for verifiable transformation rather than an impressive syllabus. These are strategic design constraints. They do not show which customer will buy, what that buyer wants, or what fee the market supports.

94. Three markets with similar technical content and different decisions

D2C: A working individual pays to improve their capability. **College partnership:** An institution buys applied curricular or training delivery that students and faculty use. **Enterprise:** A company buys team capability improvement tied to workflows, projects, or business risk. Each may involve AI, engineering practice, assessment, and teaching. But the jobs, buyers, budgets, procurement, evidence, and timing differ enough that they must be studied separately.

A fourth long-horizon stakeholder—future academic faculty, researchers, employers, donors, and regulators—matters to institutional evolution but should not be mistaken for the immediate buyer of the first product. Prestige and a campus are not prerequisites for discovering who will pay for a credible educational service today.

95. First D2C hypothesis: the working engineer facing a real assignment

One candidate segment is a software developer who already writes code, has been asked to build or maintain an AI-enabled feature, and recognizes a gap in evaluation, deployment, and engineering judgment. Their job might be: **"Become capable of making and defending reliable AI-system design decisions before I am judged on a deliverable at work."**

This segment is attractive as a *hypothesis* because its job is concrete, the founder may understand the technical challenge, and performance can potentially be assessed. But the worker may prefer internal mentorship or self-study; the organization may prohibit outside project discussion; a short review service

might suffice; the employee may not have the money or time for a long program. A rigorous discovery project has to investigate each alternative before declaring a cohort the answer.

96. The D2C job map has several distinct clusters

Potential learner situation	Dominant progress sought	Likely alternative	Main unknown
Developer assigned first AI release	Ship reliable system and defend choices	Colleague, docs, self-experimentation	Will they pay for structured expert review?
Developer planning a role transition	Credibly demonstrate new competence	MOOC, side project, formal program	Which evidence do employers accept?
Quant analyst introducing AI workflows	Build practical engineering capability	Internal team, vendor workshops	Are prerequisites and use cases sufficiently common?
Technical graduate exploring AI	Discover a career direction	Free content, campus course	Is the need urgent and funded?
Advanced practitioner seeking depth	Improve specialist judgment	Papers, peers, open source	Is founder instruction sufficiently differentiated?

These are **candidate clusters**, not observed segments. The table should determine which contrasting interviews to recruit and what evidence would refute the assumptions. They should not all be admitted into the same flagship cohort merely because they expressed interest in AI.

97. The engineer’s buying timeline, reconstructed as a hypothesis

Consider an illustrative developer assigned to a retrieval application. In week one, they try documentation. In week three, the prototype works but evaluation is unconvincing. In week four, an internal reviewer questions failure modes. The developer searches technical blogs and contacts peers. A rigorous review service becomes thinkable. But asking them to join a six-month program may miss the decision window entirely.

The key interview questions are not “Would you enjoy our AI curriculum?” They are: **What exactly failed? Who noticed? How was it repaired? What outside help was considered? What deadline governed action? What information could leave the company? What price and time were already spent?**

The product may ultimately be a compact AI-systems studio, a recurring assessed cohort, a specialized review clinic, or employer-funded team training. Discovery should preserve that uncertainty.

98. An illustrative D2C interview transcript, with inference separated

Founder: “Tell me about the last AI project where you felt uncertain about the quality of your implementation.”
Engineer: “We made a chatbot, but the product team did not trust its answers.”
Founder: “What did you try first?”
Engineer: “I watched a few tutorials and copied an evaluation script.”
Founder: “What happened when you used it?”
Engineer: “The scores looked good, but production complaints continued.”
Founder: “Who helped you investigate?”
Engineer: “A senior engineer reviewed the dataset and found leakage.”
Founder: “How long did the issue remain unresolved, and what did it affect?”
Engineer: “About a week. We delayed the release.”
Founder: “What would you do if a similar issue arose next month?”
Engineer: “Probably ask the same senior engineer, if available.”

Observation in the fictional transcript: The engineer had a concrete problem and used free material and internal expertise. **Inference:** Knowledge of evaluation design and timely review may be valuable. **Contradiction to an easy sale:** The existing internal reviewer may already satisfy the need. **Next experiment:** Interview engineers lacking that reviewer, along with managers who sponsor skills development; test a bounded, paid, non-confidential diagnostic. A good founder resists calling this interview a paid-course lead.

99. A genuine D2C readiness study

Customer understanding must include whether the learner can actually benefit. Instead of asking for self-rated AI proficiency, offer a relevant, brief task that tests prerequisite reasoning, engineering literacy, debugging, and available working time. Ask learners to show non-confidential prior work and describe the last demanding project they completed. This is not a recruitment test for prestige; it is a safety and design test for meaningful learning.

Distinguish **buyer readiness**, **learner readiness**, and **delivery readiness**. Someone may be ready to pay but not ready for advanced technical work. Someone may be highly prepared technically but unable to attend live sessions. The institution may be technically capable of teaching yet unable to promise timely reviews at proposed class size. All three must align.

100. A D2C field-research plan

Recruit a small, deliberately diverse set of people in the two strongest candidate learner situations: some who recently bought technical learning, some who declined, some who have an actual AI delivery responsibility, and some who found solutions without any course. A practical first wave might be **8–12 interviews per sharply defined situation**, adjusted as evidence emerges; these counts are planning budgets, not proof of statistical completeness. Include at least a few people outside the founder’s warm professional network.

Record for every case: trigger, desired capability, project deadline, baseline competence, existing alternative, prior learning spend, hours realistically available, payer, proof demanded, feared downside, and actual next action. Before a full cohort launch, test a **small complete paid experience** that provides real diagnostic or project-review value and permits honest outcome evaluation.

101. D2C false positives that could mislead Bayes

AI anxiety: a person worries about technological change but takes no action and will not reserve time. **Technical admiration:** an expert praises the founder's lecture yet would never enroll. **Career fantasy:** a prospect expects a guaranteed job or salary change that Bayes will not promise. **Credential demand:** the buyer primarily wants an internationally recognized qualification Bayes cannot yet provide. **Friendship purchase:** former colleagues enroll out of personal loyalty; useful for initial cash but not evidence of scalable demand. **Discount demand:** a small pilot fills only because a launch concession creates a different buyer segment.

Each is informative but not equivalent to product-market fit. Record the evidence honestly. A good institute can decline to sell to someone for whom the product is a poor fit.

102. Second market: colleges and universities are not one buyer

A college might seek industry-informed electives, hands-on labs, career preparation, faculty support, improved student engagement, accreditation-related academic development, or external differentiation. A conversation with an interested dean can reveal a worthwhile problem but not necessarily a funded program. Department chairs, faculty, administrators, students, finance staff, procurement, and placement teams may assign different value to the same proposal.

The initial research should reconstruct **the last actual curricular or external-vendor decision**. Who proposed it? Which academic body approved it? Did the college already have an elective that could be improved? What hours, labs, equipment, faculty support, timetable, and payment schedule were required? What did the institution refuse, and why?

103. A college stakeholder map

Stakeholder	Likely job	What could make them say no	Useful research evidence
Principal / leadership	Institutional quality, student appeal, strategic reputation	Reputational risk, budget, governance	Prior approved pilots and decision criteria
Department head	Curriculum fit and manageable delivery	Academic-calendar conflict, ownership	Syllabus and workload review

Stakeholder	Likely job	What could make them say no	Useful research evidence
Faculty	Teaching feasibility, professional growth	Added work, loss of autonomy, mismatch	Co-design session, lesson trial
Students	Usable capability and credible work	Weak prerequisites, time load, irrelevant topics	Diagnostic and actual practice behavior
Placement / career team	Practical preparation and employer fit	Generic training, exaggerated claims	Employer skill gaps, student performance
Finance / procurement	Compliant, predictable contract	Missing approvals or vendor requirements	Previous payment and vendor process

Treat this as a recruitment map, not a claim about a particular college's organization. Interview at least one person with actual budget or approval authority before counting the opportunity as commercially serious.

104. An illustrative college buying history

Suppose a department previously purchased seasonal placement training for graduating students. Leadership now expresses interest in integrated technical electives. The two purchases sound adjacent but differ: a seasonal program is scheduled outside ordinary curriculum; a credit-bearing elective may require committee approval, faculty coordination, assessment changes, and greater institutional accountability.

The founder should not treat an enthusiastic response to the second idea as equal to the first program's purchasing proof. Ask whether there is an existing budget, whether the program can fit an elective slot, who handles academic assessment, what external-instructor arrangements are permitted, and what constitutes acceptable student outcomes. If the decision can only occur at the next academic cycle, the revenue clock changes.

105. What Bayes should test with a college

A bounded pilot could involve a faculty-supported applied module, an assessment workshop, or a lab sequence on one defined technical outcome. The actual product should be chosen after discovery. Define the institution's responsibility, faculty involvement, student prerequisites, schedule, fee, data rights, assessment standard, and consequences if attendance is poor.

Measure not simply "students attended" but **faculty implementation load, student work quality, administrative reliability, payment timing, academic fit, and whether the buyer would repeat the engagement**. A signed memorandum without real delivery or budget is weak evidence. A paid, renewed pilot with truthful student performance reporting is far stronger.

106. College-market false positives

A college may host free guest lectures to enhance campus life without intending to pay for a long program. A dean may praise AI specialization but lack faculty approval. A placement coordinator may request mock interviews while the founder wants a deeper curricular offering. A college may seek the lowest-cost vendor rather than a rigorous partnership. A ceremonial "center of excellence" announcement may confer no operational rights, equipment, student access, or revenue.

Do not conflate courtesy, public relations, or introductory access with commercial commitment. The customer is understood only when the **actual procurement and delivery mechanism** is intelligible.

107. Third market: enterprise buyers purchase organizational change, not lectures

An enterprise team may wish to move faster with AI, reduce defects, improve analytics, comply with policy, or build internal technical capability. The learning product is a means. A manager who pays for a lecture may be seeking baseline awareness; a manager who pays for a long capability program may expect changed work behavior, better decisions, or reliable production outcomes.

Discover the **workflow before the training request**. What tasks must people perform differently? Which errors occur now? What evidence would show improvement? Who can release employees for practice? Who owns the technical environment? What skills are necessary, and what organizational impediments remain after training? If access rights or unclear responsibilities are the bottleneck, instruction alone may not solve the problem.

108. Enterprise buying may require two separate propositions

The technical manager wants employees who can build, test, and maintain an AI-enabled workflow. The L&D or HR function may want participation, standardized curricula, transparent vendor performance, and measurable development. Procurement and information security want acceptable risk. The finance sponsor wants a credible business case. These are related but distinct.

If Bayes sells "AI training," all participants might agree verbally while imagining different outcomes. A stronger specification names **one workflow capability, the target employees' starting condition, the observed business problem, the training-and-application mechanism, and the evidence of changed performance**. A small diagnostic can establish the problem and prepare approval for a larger engagement, but an engagement should remain within the founder's delivery capacity.

109. An illustrative enterprise decision map

Trigger: Engineering leadership discovers that proof-of-concept AI systems repeatedly fail release reviews. **User:** Team developers. **Champion:** Technical manager. **Economic buyer:** A department head with an upskilling budget. **Veto:** Security or procurement. **Default alternative:** Internal training or hiring a senior AI

engineer. **Potential paid experiment:** A narrowly scoped evaluation-and-skills diagnostic using synthetic or properly authorized data. **Evidence of value:** Measurable task performance, quality of an evaluation plan, implementation changes, and an acceptable cost to the team.

Main falsification possibilities: The problem is insufficient access to production data, not skill; internal mentors are cheaper; employees lack available time; the buyer will not pay external providers; procurement takes longer than the program's relevance window. A truly useful customer model describes these failure possibilities before a sales proposal is written.

110. Do not mistake an L&D budget for a willingness to buy Bayes

A company may have a large training budget but preselected vendors, annual calendars, compliance requirements, and expectations of low-cost seat-based courses. Another company may have no labeled L&D budget but an expensive operational failure and a project sponsor willing to fund a technical intervention. A large budget is not a market opportunity until its rules, owner, urgency, and fit are understood.

In enterprise interviews, ask about actual recent purchases, whether outcomes were measured, the contract approval threshold, data restrictions, and what happened after training. Observe willingness to involve the **actual budget owner**, not only enthusiastic practitioners.

111. The three-market opportunity ledger

Question	D2C working professional	College partnership	Enterprise capability
Who feels the pain?	Individual learner	Students, faculty, leadership	Engineers, managers, business unit
Who pays?	Individual or sponsor	College and/or students under agreed model	Department or L&D/operations budget
Main job	Independent technical capability	Feasible applied curriculum and student development	Better team workflow capability
Default substitute	Self-study, internal peer, competing course	Internal faculty, existing vendor, no change	Internal coaching, vendor, consultant, hiring
Main buyer uncertainty	Time, credibility, actual learning gain	Scheduling, academic fit, delivery reliability	Business effect, procurement, security
Plausible first proof	Small paid review / studio	Paid academic pilot	Paid skill/workflow diagnostic

Question	D2C working professional	College partnership	Enterprise capability
Main founder bottleneck	Live review and accountability	Calendar coordination and approvals	Sales cycle and customization
Stronger demand evidence	Paid qualified learner who does the work	Approved paid contract and delivered pilot	Budgeted pilot with owner and acceptance criteria

This table is a **hypothesis ledger**, not a conclusion that all three markets are attractive or that any particular offer will sell. Its purpose is to prevent the founder from mixing unlike observations into one imaginary "AI education demand" statistic.

112. Which segment should Bayes interview before it builds?

Chapter 5 identified opportunity spaces; this chapter seeks the decision mechanism inside them. A sensible first research cycle would choose **one narrow learner situation and one narrow institutional situation**, rather than launching broad campaigns across three sectors. Compare the information gain per founder-hour with the commercial stakes. The order should be updated based on reachable decision-makers, not on an unvalidated grand narrative.

A candidate first wave: working engineers with recent AI delivery responsibility and recent learning expenditures; college academic owners who recently procured or rejected applied electives; enterprise technical managers who recently sponsored externally delivered AI work. Interview actual nonbuyers alongside buyers in each tested domain. Only extend to more markets when the initial evidence is discriminating, not simply encouraging.

113. What would actually change the Bayes course design?

If working engineers mainly fail at architecture review and cannot spare weekly lectures, a long live-heavy course may be poorly aligned. If they struggle to maintain practice and value accountability, a sustained cohort becomes more plausible. If they want an independently recognized degree, Bayes's current non-degree offering may not satisfy the job. If the payer is an employer, a team-based curriculum might be more viable than individual tuition. If beginners cannot complete prerequisite tasks, the institute must narrow admissions or offer a separate preparatory path.

The founder should enter research with at least three viable product hypotheses and allow evidence to change the choice. **Customer understanding becomes strategically valuable when it changes what the founder would otherwise build.**

114. How to test student capability demand without promising career

outcomes

Design a transparent demonstration of a real engineering task: evaluate a small AI system, identify leakage or unreliable metrics, propose an improvement, and defend trade-offs. Use a rubric aligned with the kind of capability the course would eventually teach. Invite relevant prospective learners to attempt the task in an authorized setting.

Observe whether they value feedback, whether their errors are systematic, how much time is required, whether the review improves subsequent work, and whether they would pay for a **complete bounded experience** of this kind. The test does not establish that a six-month cohort causes promotions or jobs. It establishes whether a particular technical struggle is real and whether expert intervention may be useful.

115. A Bayes-specific customer-evidence dashboard

Track four separate groups of indicators. **Problem evidence:** number of recent cases with consequential failure; actual workarounds; deadlines; prior spending; identifiable payer. **Commitment evidence:** paid pilots, completed diagnostics, signed terms, budget approval, repeat invitations. **Transformation evidence:** baseline and unseen endline performance, follow-through, reviewer agreement, attrition, learner effort. **Operating evidence:** founder hours, feedback capacity, acquisition burden, collection timing, and cost per defensible outcome.

Do not combine them into a single customer-understanding score. A program can find an urgent problem but fail to deliver; a successful technical workshop can fail to attract buyers; a profitable contract can damage the mission if the participants gain nothing. The separate indicators reveal which mechanism needs improvement.

116. A decision Bayes should be willing to make after research

Suppose discovery shows high stated interest among general AI enthusiasts but low willingness to commit time or money. A smaller subset of working engineers has concrete problems, pays for narrow expert reviews, and completes difficult assignments. Colleges express interest but face long approvals. Enterprise managers have meaningful budgets but require customized delivery and founder attention beyond current capacity.

If those were actual observations—not facts asserted here—the initial conclusion might be to focus on a narrow paid engineering studio, defer broader college sales, and selectively test enterprise diagnostics. But another research result could rationally reverse the priority. The book's method should **make the decision contingent on evidence**, not force reality to validate the founder's favorite first product.

117. Customer knowledge is an institutional asset only when it survives the founder

An early institute may benefit from the founder knowing every learner personally. As it grows, that knowledge must become a **consent-aware, decision-useful system**: learner starting conditions, recurring

misconceptions, reasons for withdrawal, employer needs, partner constraints, and evidence of interventions. This is not permission to accumulate intrusive personal profiles or indefinitely retain student data.

The valuable asset is a reliable ability to *identify, serve, and learn from recurring customer situations*. A database of stale contacts or broad student demographics is not the same thing. Document the decisions the evidence improves, update it after delivery, and remove data when no longer justified.

Part X — From Customer Knowledge to Better Founder Decisions

118. Customer understanding should produce a causal hypothesis

After research, write a statement that can be wrong:

> **When [trigger] occurs, [specific buyer] struggles to achieve [progress] using [actual alternative] because [observable deficiency]. They already incur [time, money, failure, or risk] and can authorize [specific commitment]. They would plausibly choose [new mechanism] if [evidence or condition] is satisfied. The theory fails if [counterexample or observed refusal].**

If the sentence contains adjectives like “innovative,” “future-ready,” or “world-class” instead of actual behavior, the research is not complete. The statement is not a product specification. It is the causal theory that product design will later test.

119. Separate problem–customer fit from solution–customer fit

A founder can establish that a group shares a consequential problem without proving that a specific course, app, service, or price solves it economically. The first is **problem–customer fit**: the identified group repeatedly experiences the problem and has the means and motivation to address it. The second is **solution–customer fit**: a proposed mechanism improves the relevant outcome in a form the customer can adopt and will fund. Full product–market fit requires stronger evidence of repeatable demand and delivery, which Chapter 9 will analyze.

Do not use these as official standardized stages or guarantee that discovery is linear. In practice, proposing a solution can reveal a previously hidden problem, and a paid pilot can change the target segment. The distinction remains useful because it tells the founder *which belief has been tested*.

120. Rank unknowns by their ability to change your decision

A founder can collect endless interesting details. Instead ask which uncertainty could reverse the investment decision: Is there an urgent buyer? Can they authorize payment? Is the existing workaround genuinely costly?

Does the target group have the prerequisites? Can the product deliver within founder capacity? Does the customer need a credential that the business cannot offer?

List the unknowns and rank them by **decision importance × current uncertainty × affordability of a meaningful test**. This is an informal prioritization heuristic; do not multiply arbitrary scales to create false scientific precision. Test fatal uncertainties early, especially those involving access, payment, legal feasibility, and operational impossibility.

121. Look for the buyer's actual selection criteria

A founder should understand the criteria **used**, not only the criteria a respondent says ought to be used. Ask for the final two options considered, why each was rejected, which evidence tipped the choice, and what compromise was accepted. The result may surprise you: the enterprise picked a weaker tool for security approval, the student chose an ordinary course for schedule flexibility, the retailer chose a supplier for stable replenishment.

This becomes the foundation for differentiation in Chapter 8. The most useful selection criteria are the ones that **change the final decision**, especially when they are expensive for incumbents to satisfy. Always inspect exceptions; customers can misremember their reasons and contexts change.

122. Understand the counterfactual, not just the customer's complaint

If your proposed service vanished, what would the customer do next? If the answer is "ask a colleague, who would probably solve it in thirty minutes," you may have a weak paid opportunity. If the answer is "delay a release for three weeks and spend significant senior engineering time," the problem could be valuable—assuming a buyer recognizes the cost and an external provider is allowed to help.

A complete business case compares **outcomes with the new solution** against **the best feasible alternative**, accounting for implementation and customer effort. This counterfactual discipline prevents founders from claiming the entire value of a customer's success as if it would not have happened otherwise.

123. Build a customer theory that predicts a particular refusal

A useful model should explain not only why buyers accept but why they reject. For example: "Working engineers with high urgency, inadequate internal support, prerequisite knowledge, and available time will consider supervised practice; engineers with strong internal mentors will usually decline; those lacking prerequisites will not be served responsibly by the advanced offering."

Before the next field interaction, make a small forecast about which of these types will commit, what objection will dominate, and which alternative they will choose. Then record what happened. This connects customer understanding to the forecasting and Bayesian decision discipline from Chapter 4. **Predictive failures are valuable because they expose missing mechanisms.**

124. The five customer models every founder should be able to explain

You should be able to reconstruct **the problem model** (trigger, stakes, workaround); **the choice model** (alternatives and decisive trade-offs); **the authority model** (who can approve and pay); **the adoption model** (what must happen after purchase to create value); and **the economic model** (what the buyer sacrifices and the firm can retain). These models may exist in one notebook rather than five corporate documents.

If you can explain only the problem, you are doing empathy without business design. If you can explain only the choice, you may be copying marketing without understanding value. If you can explain only the economic model, you may be pricing transactions that will not happen. Their interaction is what makes customer understanding an entrepreneurial capability.

125. Customer discovery has a stopping rule: action, not omniscience

The goal is not perfect knowledge of every customer. It is enough decision-relevant understanding to select the next **bounded, reversible, ethically deliverable** experiment. A founder should stop interviewing temporarily when additional conversations are unlikely to change the design of that experiment and when a real behavior test is feasible.

Then run the test, observe delivery, and return to discovery with sharper questions. Repeated interviews without action can become avoidance. Building an entire business before talking to buyers is the opposite error. The discipline is **learn enough to act responsibly; act in a way that reveals more; update the theory**.

Appendices — The Customer-Understanding Field Manual

The appendices are designed to be used repeatedly. They are not a replacement for judgment, direct observation, statistical design where appropriate, or responsible handling of people and data. Their value lies in making the founder's assumptions, evidence, and next decisions visible.

Appendix A — The one-page customer decision dossier

Create **one** dossier for each materially different buyer and job. Do not average a learner, dean, and engineering director into a single row.

Field	Write the evidence-backed answer
Customer situation	Who, doing what, in which context?
Trigger	What specific event made this problem salient?

Field	Write the evidence-backed answer
Present job	What progress are they trying to make?
Functional / emotional / social dimensions	What changes and who notices?
Consequences	What happens when the job is done badly or not at all?
Frequency and urgency	Recent incidents and deadlines
Current method	Actual product, person, workaround, or nonconsumption
Best feasible alternative	What would the customer do if we did not exist?
Existing sacrifice	Money, time, risk, coordination, embarrassment
Desired outcome	Evidence that the customer would regard as success
Proposed mechanism	A candidate service/product, explicitly a hypothesis
Likely user, buyer, payer	Exact roles, names only with legitimate need and consent
Veto and approval process	What can block purchase?
Relevant buying calendar	Budget and implementation timing
Trust threshold	What must be verifiable before commitment?
Current evidence	Interviews, observations, artifacts, previous transactions
Contradictions	Cases that resist the hypothesis
Next costly test	Specific real action that would discriminate alternatives
Confidence	Observed / reported / inferred / untested
Decision	Pursue, revise, defer, reject—with reason

The discipline: Every important field should be traceable to at least one source, not merely a founder's intuition. One respondent can justify a hypothesis; repeated independent observations and costly commitments warrant stronger confidence.

Appendix B — The forty-five-minute buyer-history interview guide

Preparation. Confirm the respondent has recently experienced or decided about the specific job. Note their role and the route by which they were recruited. Prepare open questions and an alternative hypothesis.

Establish privacy and recording consent.

Opening. “I’m trying to understand how people handle [specific situation]. I’m not asking you to validate my product. Would you walk me through an actual recent case?”

Trigger. “When did you first notice this problem?” “What had just changed?” “Why did it matter at that point?” “What would have happened if nothing changed?”

Existing behavior. “What did you do first?” “Which tools, people, or services did you use?” “Could you describe the steps?” “Which part took the most time or caused the most risk?”

Alternatives. “Did you consider another approach?” “How did you discover it?” “Who recommended it?” “Why wasn’t it chosen?” “What still works well about your current method?”

Purchasing. “Who had to agree?” “Where did the budget come from?” “What was the last comparable thing you paid for?” “What documents or approvals were needed?” “What nearly stopped the decision?”

Consequences. “What happened after you chose?” “What did it actually change?” “What disappointed you?” “Would you make the same choice today?”

Close. “What important part of the decision did I fail to ask about?” “Is there someone who made a different choice in a similar situation whom I could speak with, if they would welcome the introduction?”

Afterward. Within the same day, record the timeline, observations, exact respondent words where needed, the researcher’s inferences, contradiction cases, and the next hypothesis. Do not add the respondent to marketing communications without appropriate permission.

Appendix C — Three specialized interview paths for Bayes Institute

C1. Working engineer or technical analyst

1. What real AI or engineering task has been most difficult during the past three months?
2. When did that task first become your responsibility, and when was its deadline?
3. What could you already do independently before attempting it?
4. Which parts failed, and how did you discover the failure?
5. Which help did you seek—internal mentors, code assistants, docs, communities, paid courses, consultants?
6. What did each alternative do well or poorly?
7. Did you or your employer pay for any help? What was authorized and what was not?
8. What would you need to be able to do next time without external help?
9. How many hours did you *actually* spend learning in recent weeks? What blocked more?
10. What would make an unknown institution credible enough for a substantial commitment?
11. Which kind of evidence matters more to you: a sample lesson, review, independent assessment, prior graduates, employer recognition, or something else? Ask for examples rather than a ranking alone.
12. What do you currently believe a course cannot give you?

C2. College academic leader or department owner

1. Tell me about the most recent externally delivered curricular or skills program that was proposed.
2. What problem initiated the discussion, and who raised it?
3. What was the formal process from proposal to approval or rejection?
4. Which academic, legal, budget, timetable, and assessment constraints mattered?
5. How were faculty involved, and what did they accept or resist?
6. What prerequisites did students actually have? How were differences handled?
7. What provider alternatives were evaluated, and why was one selected?
8. How were delivery success and student learning checked?
9. What was paid, when, and by which entity? What approvals and invoices were needed?
10. If repeating the program, what would you change and who would authorize that change?
11. Which upcoming academic date creates the next genuine purchase opportunity?
12. Would a narrowly scoped paid pilot be considered under existing policy? What exact steps would be required?

C3. Enterprise engineering or capability buyer

1. Which AI-related work task is materially underperforming today?
2. How do you know? What baseline failures, delays, or costs are visible?
3. What proportion of the difficulty is people capability versus data, access, architecture, process, or incentives?
4. What have internal experts or past vendors already tried?
5. Who would need to behave differently after any training?
6. What time can employees realistically be released for application and feedback?
7. What does success look like at the workflow level, not only course completion?
8. Who owns budget, technical evaluation, security review, and vendor approval?
9. How was the last comparable outside engagement procured and paid?
10. What would make a bounded diagnostic worth paying for before a longer program?
11. What data or code may be accessed and under whose authorization?
12. What evidence would support renewal or expansion of an initial engagement?

None of these are questions to fire rapidly in sequence. They are a bank from which to investigate *one actual episode* deeply. For delicate corporate information, accept refusal and work with authorized, non-confidential scenarios.

Appendix D — The observation and artifact checklist

When the founder can safely observe a real workflow, capture: task objective; context; starting state; sequence of actions; tools and systems used; delays; handoffs; errors; help sought; workarounds; approvals; rework; final

output; and how the participant judges success. Estimate time from a clock where possible rather than recalling it later. Note what the observer's presence might have changed.

For learning, observe a realistic independent technical task and distinguish conceptual misunderstanding, tool friction, missing prerequisites, time pressure, and evaluation anxiety. For colleges, observe timetable and grading workflow where permitted. For enterprise, study non-confidential workflow diagrams or appropriately authorized synthetic tasks. **Do not request private learner records, proprietary client data, or production credentials merely to make discovery look rigorous.**

Appendix E — An opportunity/problem evidence register

Hypothesis	Supporting observation	Contradicting observation	Strength	Next decisive test
Engineers need technical feedback more than content	Recent developer relied on senior code review to fix leakage	Several developers have sufficient in-house review	Preliminary, context-specific	Paid diagnostic with engineers lacking internal review
Colleges will buy integrated electives	Some leaders express interest in applied curricula	Approval route and budget remain unclear	Weak commercial evidence	Interview last purchasing committee; seek scoped paid pilot
Enterprise buyers need measurable AI capability	Manager reports recurring deployment errors	May be data/process issue, not training	Problem not yet diagnosed	Workflow + skills audit with appropriate sponsor
Working professionals can sustain 6-month course	Interviewees say long-form learning is attractive	Calendars show limited recurring study time	Contradictory	Short assessed sprint; observe actual effort

Do not mechanically compute a confidence percentage from interview mentions. Track how observations were recruited, whether they are independent, and whether they discriminate between rival explanations.

Appendix F — The acute-problem interrogation

Take one suspected opportunity and answer the following in writing:

1. **Actual event:** What happened in the last month or quarter, and to whom?
2. **Stakes:** What concrete loss, obstacle, or delay followed?

3. **Frequency:** How often does an event like this occur in the relevant segment?
4. **Workaround:** What do people use or do today, and what does it cost?
5. **Authority:** Who can authorize a change and who might veto?
6. **Urgency:** What happens if the situation persists for another relevant period?
7. **Alternative:** What good-enough solution would the customer choose without us?
8. **Evidence:** What actual behavior supports each answer?
9. **Feasibility:** Can our organization improve the situation under its actual constraints?
10. **First exchange:** What smallest complete paid commitment could test this?
11. **Disproof:** What outcome would make us abandon or redraw the segment?

A problem with massive emotional significance and no lawful or commercially practical method of serving it may be a worthy mission project yet a weak bootstrapped entry product. The tool is intended to make that distinction explicit.

Appendix G — Reconstruct a purchase as a timeline

```
Customer / role / recruitment source:
Actual job / context:

T-?  First background dissatisfaction:
T-?  Event that made the situation urgent:
T-?  First active search or conversation:
T-?  Alternatives considered, including doing nothing:
T-?  First convincing evidence:
T-?  Main anxiety, objection, or veto:
T-?  Budget owner and approval process:
T0   Commitment: payment, contract, time, or refusal:
T+?  Activation / first meaningful use:
T+?  Measured result / disappointment:
T+?  Renewal, recommendation, regret, or abandonment:

Facts personally observed:
Facts reported by customer:
Artifacts reviewed lawfully:
Founder's interpretation:
Strongest rival interpretation:
Most important unresolved uncertainty:
Next experiment:
```

This timeline is especially useful when a respondent says they “decided immediately.” Often the conscious decision appears sudden only because earlier problem recognition, recommendations, and search happened invisibly over weeks or months. In other cases the decision truly is impulsive; do not force every purchase into a long story.

Appendix H — An illustrative comparison of two segments

Suppose a founder finds **two hypothetical clusters** during research. Cluster A comprises 20 general learners expressing interest in AI, of whom 16 say a substantial program sounds appealing but only 2 have recently paid for anything similar and most report no study time. Cluster B comprises 12 working engineers with a concrete delivery assignment, of whom 8 have recently spent significant time or money resolving AI implementation problems, 6 report inadequate review, and 3 accept an actual paid technical diagnostic at a specified fee.

This does **not** establish that Cluster B is a large or statistically superior market: the samples were small, recruited differently, and may have selection bias. It does suggest a better next experiment for the founder. The diagnostic purchase shows a certain solution can attract some real commitment in B; it does not validate a long course or a particular future conversion rate.

A disciplined conclusion might read: *"In this small, nonrepresentative sample, actual problem resolution and limited paid engagement appeared more common in the delivery-responsibility cluster. We will test a second independent recruitment source, measure fulfillment effort, and compare the paid diagnostic against the most relevant free or internal alternative. We will not extrapolate demand to all developers."*

This is the kind of modest but decision-changing inference a founder should learn to make.

Appendix I — The first 60 days of customer discovery

Days 1–7: Frame the uncertainty

Write down no more than three candidate customer situations. For each, name the trigger, job, current alternative, payer, purchasing constraints, proposed first value, and main reason the market may not exist. Build an interview recruitment list through legitimate sources. Do not spend the week polishing a product site.

Deliverable: Three competing customer hypotheses, each with a strong falsification question.

Days 8–18: Reconstruct behavior

Conduct a first wave of focused interviews and, where possible, observe non-confidential workflows. Include recent buyers and nonbuyers. Record comparable facts, not merely quotes. Follow unexpected cases. At the end of each day, identify what you wrongly assumed.

Deliverable: Initial decision histories, alternative map, contradiction register.

Days 19–28: Challenge the emerging explanation

Recruit people outside the original circle. Seek someone who solved the same problem without buying, someone who bought a different solution, and someone whose purchasing process failed. Interview the actual budget owner where user and payer differ. Compare apparent segments by decision behavior.

Deliverable: Revised customer decision dossiers and a shortlist of unresolved bottlenecks.

Days 29–40: Offer a complete small experiment

Construct a clearly scoped, responsibly deliverable diagnostic, task review, technical workshop, pilot, sample, or software workflow—depending on the industry. Specify the real price or commitment and the customer's obligations. Predefine what success and failure mean. Avoid presenting the product as proven when it is experimental.

Deliverable: Real behavior data: qualified invitations, acceptance, payment, participation, outcome, cost and objections.

Days 41–50: Examine the full experience

Observe actual use and nonuse. Ask purchasers whether the experience solved the job and what they would now do differently. Track refunds, dropouts, delays, support, buyer regret, and time spent. Reinterview appropriate nonbuyers without pressure.

Deliverable: Delivery evidence and corrected job, alternative, and economic hypotheses.

Days 51–60: Decide what deserves the next stage

Compare the hypotheses using **hard constraints** (legal and ethical feasibility, budget access, founder capacity, demonstrable customer problem) before debating growth potential. Choose one focused customer situation for the next serious experiment, identify what remains unknown, and explicitly park attractive but unsupported ideas.

Deliverable: Customer-understanding memo, one proposed beachhead, and evidence needed for Chapter 7.

Timing caveat: A university procurement cycle or a six-month learning transformation cannot be validated in sixty days. The exercise sets a research and decision cadence, not a deadline for claiming facts that have not matured.

Appendix J — Weekly customer-learning meeting

Even a solo founder can reserve 30 minutes to answer: **What did customers do, not just say? What surprised us? Which current alternative remains strongest? Which purchasing role or authority is still unverified? What did we learn from a nonbuyer? Which evidence is contaminated by our pitch or recruitment method? Which hypothesis will change our next product or distribution choice?**

When the business grows, include sales, instruction/delivery, support, and product. Reward the person who finds a contradiction, not only the person who brings a promising lead. A customer-learning meeting that never changes decisions is bureaucracy, not learning.

Appendix K — Common mistakes and their corrections

Tempting founder habit	Why it fails	More disciplined practice
Interview only friends	Courtesy and network selection	Recruit genuine buyers, nonbuyers, strangers where appropriate
Ask whether the idea is good	Evaluates social politeness and imagination	Investigate recent real decisions
Treat a feature request as a job	Customer proposes means, not necessarily best ends	Ask what breaks without the feature
Count complaints as buying intent	Dissatisfaction can coexist with inertia	Study costly remedies and switching thresholds
Use “all 25-year-olds” as a segment	Biographies do not determine buying occasion	Segment by trigger, job, alternative and capacity
Believe the champion controls budget	Organizational approval is distributed	Identify signer, payer and vetoes
Use five usability tests to estimate market size	Method targets usability issues	Apply suitable market sampling and behavioral tests
Hide uncomfortable findings	Produces a self-confirming story	Maintain a contradiction register
Assume payment proves quality	Customers can regret purchases	Measure activation, outcome, repeat and regret
Build features from survey popularity	Preferences are context-dependent and often hypothetical	Compare actual trade-offs, paid tests and usage
Call a free workshop a validated premium course	Commitment levels and products differ	Test the real next purchase separately
Ignore founder hours	Manual acquisition and delivery are not free	Record economic and cash cost separately
Ask AI to invent customer data	Plausible narratives are not observations	Use AI to organize consented evidence, never fabricate it
Collect sensitive data “just in case”	Privacy and trust harms	Minimize, protect, purpose-limit and delete appropriately

Appendix L — Further reading and research foundations

This chapter synthesizes research, conceptual frameworks and field practice. The items below are not interchangeable: controlled experimental findings explain certain decision tendencies; qualitative methods help discover mechanisms; entrepreneurial books recommend practices; industry reports describe particular purchasing settings. The chapter's worksheets and Bayes cases are proposed tools, not directly validated instruments. Any local legal, privacy, education, tax or accreditation requirements should be independently verified before implementing a program or data-collection process.

[R1] Herbert A. Simon (1955), "A Behavioral Model of Rational Choice." *Quarterly Journal of Economics.* Foundational work on bounded rationality and decision procedures under limited cognition. Journal.

[R2] George A. Akerlof (1970), "The Market for 'Lemons': Quality Uncertainty and the Market Mechanism." *Quarterly Journal of Economics.* Why hidden quality changes market behavior; important for education and professional services. Journal.

[R3] Daniel Kahneman and Amos Tversky (1979), "Prospect Theory: An Analysis of Decision under Risk." *Econometrica.* Reference dependence, losses and decision weights in risky choice. Does not imply universal manipulation techniques. Abstract.

[R4] William Samuelson and Richard Zeckhauser (1988), "Status Quo Bias in Decision Making." *Journal of Risk and Uncertainty.* Experiments and field settings illustrating persistence with existing options. Journal.

[R5] Brigitte C. Madrian and Dennis F. Shea (2001), "The Power of Suggestion: Inertia in 401(k) Participation and Savings Behavior." *Quarterly Journal of Economics.* Effects of default arrangements in a specific retirement-plan setting. NBER version.

[R6] Clayton M. Christensen, Taddy Hall, Karen Dillon, and David S. Duncan (2016), "Know Your Customers' 'Jobs to Be Done.'" *Harvard Business Review.* A formative statement of the circumstances-and-progress lens. Original.

[R7] Christensen Institute, "Jobs to Be Done Theory." Institutional explanation of functional, social, and emotional progress, including circumstance and decision forces. Overview.

[R8] Daniel Yankelovich and David Meer (2006), "Rediscovering Market Segmentation." *Harvard Business Review.* On segmentation's connection to actual purchasing behavior and decisions instead of decorative profiles. Original.

[R9] Steve Blank and Bob Dorf, *The Startup Owner's Manual*, and Steve Blank's customer-development teaching. Practitioner framework for testing business assumptions with customers and separating discovery from validation. Kauffman introduction; Manifesto.

[R10] Jakob Nielsen (2000), "Why You Only Need to Test with 5 Users," and subsequent Nielsen Norman Group guidance. About small *usability* studies and iterative discovery of usability issues—not evidence that five interviews prove market demand. Original; Clarification.

[R11] Nielsen Norman Group (2020), "Contextual Inquiry: Inspire Design by Observing and Interviewing Users in Their Context." An accessible explanation of observational research in users' actual

work settings. Guide.

[R12] US Census Bureau, "Questionnaire Testing and Evaluation Methods for Censuses and Surveys." Authoritative guidance on pretesting questions, cognitive interviews, comprehension and order effects. Methods.

[R13] John A. List and Craig A. Gallet (2001), "What Experimental Protocol Influence Disparities Between Actual and Hypothetical Stated Values?" *Environmental and Resource Economics.* Meta-analysis on hypothetical bias under particular experimental valuation methods; the estimate is not a valid plug-in correction for startup surveys. Journal.

[R14] Gartner, "The B2B Buying Journey," and related research. Non-linear B2B purchasing and six buying jobs. Industry-research description; do not assume every purchase uses the same structure. Guide; 2019 report.

[R15] Icek Ajzen (1991), "The Theory of Planned Behavior." *Organizational Behavior and Human Decision Processes.* A structured account of how attitudes, norms and perceived control relate to behavioral intentions and action. Particularly helpful for understanding why positive attitudes do not automatically produce behavior. Journal.

[R16] K. Anders Ericsson and Herbert A. Simon (1993), *Protocol Analysis: Verbal Reports as Data*. A methodological foundation for using verbal reports, with important caveats about what verbalizations do and do not reveal. MIT Press.

[R17] Rob Fitzpatrick, *The Mom Test*. A practical guide to avoiding leading interviews, compliments and hypothetical customer validation. It is a practitioner method, not a statistical guarantee. Publisher.

[R18] Sarah Lichtenstein and Paul Slovic (1971), "Reversals of Preference Between Bids and Choices in Gambling Decisions." *Journal of Experimental Psychology.* Classic evidence that elicitation method can affect expressed preferences in risky decisions; it does not directly estimate demand for any particular product. Research record.

Suggested order for a first-time founder

Start with Christensen and the Christensen Institute for the jobs lens; read Blank and Fitzpatrick for field habits; read Yankelovich and Meer for decision-relevant segmentation; read Simon, Kahneman–Tversky, Samuelson–Zeckhauser, and Ajzen for decision psychology; use Census and contextual-inquiry methods when designing research; consult Gartner when studying complex organizational purchases. **Then make real observations and purchases the judge of your theory.** No amount of reading can substitute for that part.

The chapter's essential principles

First, people do not buy categories; they make situated choices to improve some aspect of their lives or work. The relevant unit of research is often a recent decision or struggle, not a demographic label.

Second, need, interest, pain, and paid demand are different observations. The practical job is to discover the bridge from one to the other: urgency, alternatives, resources, trust, and authority.

Third, every customer has an alternative, even when it is doing nothing. Understanding why the alternative persists can teach more than cataloging its flaws.

Fourth, customers have constraints that founders frequently underestimate. Time, risk, identity, learning burden, procurement and existing relationships are as real as the sticker price.

Fifth, the strongest early research reconstructs actual behavior. Interviews reveal hypotheses; observation and artifacts correct them; paid tests and subsequent use reveal additional truths.

Sixth, a business buyer is not necessarily a single person. Users, champions, payers, approvers and veto holders create a joint purchasing problem.

Seventh, a useful segment is one that changes business choices. An elegant persona that cannot change product, price, access or service is decoration.

Eighth, a good customer theory predicts refusals as well as purchases. Contradictory observations are not distractions. They are opportunities to revise a faulty causal model.

Ninth, the founder's job is not to convince everyone. It is to discover whom the business can serve unusually well, at an acceptable cost and risk, and make the purchase understandable and fair.

Tenth, understanding is an ongoing capability. Every new customer, changed technology, rejected proposal, delivery failure and successful alternative should update the organization's view of the market.

Final thought — The moment the founder begins to see

A beginning entrepreneur often sits alone imagining a customer. They imagine what the person wants, what would delight them, how much they will pay, and why the founder's product will be better. The imagined customer is wonderfully cooperative: they recognize their problem, understand the offer, trust the provider, have the required money and time, and purchase for the reasons the founder hoped.

The real customer has a life. They are busy, skeptical, sometimes inconsistent, often satisfied enough, and constrained by obligations the founder cannot see from a spreadsheet. They may want an outcome without wanting the product, love the product without affording it, afford it without needing it, need it without being allowed to buy it, or buy it without using it. Some will benefit most from a solution the founder never considered.

Entrepreneurship begins to become serious when the founder stops trying to make the real customer resemble the imagined one.

The purpose of customer understanding is not to become better at telling people what they ought to want. It is to acquire the humility, observational skill, causal imagination, and commercial discipline to ask:

> **What is this person actually trying to accomplish? What do they do today? Why does that approach persist? What would have to change for a different choice to be worth making? And how**

> **can we find out without fooling ourselves—or them?**

If Chapter 5 taught us to find a market worth entering, Chapter 6 teaches us to recognize the actual decision taking place inside it. Chapter 7 can now ask a sharper question: **Among all the people who might benefit, which narrowly defined group should we serve first—so exceptionally well that they give us the evidence and resources to earn a broader future?**

End of Chapter 6.